Space-time correlations and anisotropy of Lagrangian particle dispersion in magnetohydrodynamic turbulence

Masterarbeit

von

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Köln, 8. September 2017

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Abstract

To decipher the structure of the turbulent motions in a magnetohydrodynamic (MHD) fluid novel analysis methods are applied to a spatially and temporally resolved MHD simulation. The presence of the magnetic field and MHD waves add to the pool of already complex interactions found in the turbulence of neutral fluids. Anisotropy in the decay of eddies is a known property in MHD, where the mean magnetic field acts as a reference direction. How this anisotropy infers the actual flow of particles is unknown. We analyze the relative displacement of traced particles in a cloud using the convex hull of the cloud, the principal components of the displacements and the cloud's shape compared to a spheroid. These methods reveal the anisotropy and shape of the relative displacement and allow the comparison with characteristic directions, e.g. the mean velocity or the mean magnetic field. Additionally, in the Eulerian frame of reference, space-time correlations are applied to the velocity, magnetic field and external forcing to investigate any present coupling. This space-time correlation function is extended to the scale-dependent correlation function, allowing the analysis of decorrelation effects for fluctuations on a specific scale in MHD.

Zusammenfassung

Zur Entschlüsselung der Struktur turbulenten Strömungen in einer magnetohydrodynamischen (MHD) Flüssigkeit werden neuartige Analysemethoden auf eine räumlich und zeitlich aufgelösten MHD-Simulation angewandt. Die Präsenz des Magnetfeldes und der MHD-Wellen führt zusätzliche, neben den in der Turbulenz neutraler Flüssigkeiten bereits vorhandenen, komplexe Wechselwirkungen ein. Die Anisotropie im Wirbelzerfall ist ein bekanntes Phänomen der MHD, bei der das mittlere Magnetfeld als Referenzrichtung dient. Wie diese Anisotropie die tatsächliche Bewegung von Teilchen beeinflusst, ist unbekannt. Wir analysieren die relativen Abstände der Teilchen in einer Wolke mit Hilfe der konvexen Hülle der Wolke, den Hauptkomponenten der Abstände und die Wolkenform im Vergleich zu einem Sphäroiden. Diese Methoden zeigen die Anisotropie und die Form der Wolken auf und erlauben den Vergleich mit charakteristischen Richtungen, wie z. B. der mittleren Geschwindigkeit oder dem mittleren Magnetfeld. Zusätzlich werden im Eulerschen Bezugssystem Raum-Zeit-Korrelationen auf die Geschwindigkeit, das Magnetfeld und die äußeren Kräfte angewandt, um jede vorhandene Kopplung zu untersuchen. Diese Raum-Zeit-Korrelationsfunktion wird auf die skalenabhängige Korrelationsfunktion erweitert. Damit ist die Isolierung von Effekten, die zur Dekorrelation von Fluktuationen von einer bestimmten Skala in der MHD führen, möglich.

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1 Introduction

Statistics have been the cornerstone of research on the properties of turbulent flows (Taylor, 1922, 1935). The revolutionizing work of Kolmogorov in the year 1941 has led to the most famous result in turbulence research, the $E(k) \sim k^{-5/3}$ wave number energy density spectrum, where he uses the assumptions of restored symmetries in turbulent chaotic motions in a statistical sense (Frisch, 1995). The universality of this power-law is now well known and it is found in many experimental and numerical studies (Nelkin, 1994). Similarly, theories on the wave number energy density spectrum have been derived in magnetohydrodynamics (MHD), the description of electrically conducting fluids (Iroshnikov, 1964; Kraichnan, 1964; Matthaeus et al., 1990; Goldreich and Sridhar, 1995; Boldyrev, 2006). Here, the energy density distribution is found to be anisotropic with respect to the local mean magnetic field (Cho and Vishniac, 2000; Maron and Goldreich, 2001; Cho et al., 2002; Cho and Lazarian, 2004; Horbury et al., 2008; Beresnyak and Lazarian, 2009; Cho and Lazarian, 2009; Chen et al., 2011; Horbury et al., 2012; Oughton et al., 2015; Gerick et al., 2017). These spectra are directly related to the flow field at a distinct snapshot in time. No temporal dynamics are considered or needed. The evolution of turbulence with time has been a niche field of research. The major aim of the studies considering the time domain were the validation of the frozen flow approximation, which allows for temporal observations at one point in space to be regarded as a spatial observation at one point in time (Taylor, 1938; Favre et al., 1961; Favre, 1965; He and Zhang, 2006; He et al., 2017).

The investigation of the temporal dynamics of MHD turbulence and the actual flow field in space and time is the aim of this thesis. The presence of the magnetic field introduces waves and anisotropy to the system. These additional features lead to new possible interactions and mechanisms in the temporal evolution of MHD flows. The decorrelation of the flow fields can be analyzed and compared to the characteristic times of these new mechanisms to determine their significance (Servidio et al., 2011; Lugones et al., 2016; Perri et al., 2017). When analyzing flow fields in the time domain, two frameworks are generally distinguished. Looking at fluctuations in time at a point in a fixed frame is known as the Eulerian framework. The second is the Lagrangian framework where the flow field is observed along a particle's trajectory through the flow. The major focus has been on the Eulerian framework, because most theory is build on it and the observation is easier. In the Lagrangian frame there have been some approaches in HD

turbulence, where laboratory experiments allow the tracing of particles before numerical simulations were available (Durst et al., 1959; Deardorff and Peskin, 1970; Shen and Yeung, 1997). For the MHD case tracing of particles or pseudo-particles is very difficult. Therefore only very recently the analysis of particle trajectories in MHD flows has been available by the use of high resolution simulations (Busse et al., 2007). Still, to this day a sophisticated computer infrastructure is needed to store the full simulation in spatial and temporal domain, necessary to trace particles in a simulation (Li et al., 2008). The dispersion of the particles in the flow has deep connections to the transfer of energy of the fluctuations or eddies in the flow. The rate and directional preferences of the particle dispersion are directly governed by the flow characteristics and are expected to give insight into the structure of energy transport in turbulence (Richardson, 1926; Batchelor, 1950; Busse et al., 2007; Servidio et al., 2016).

Besides the coupling between the flow fields and the external forcing, we want to investigate their temporal decorrelation and the anisotropic dispersion of particles in the simulation. In Chapter 2 we introduce the basic physical concepts relevant in HD and MHD turbulence. Chapter 3 introduces the MHD simulation this work is based on. Concepts and methods of statistical analysis are introduced in Chapter 4. The Chapters 5 and 6 present the results of the introduced methods applied to the simulation data under consideration. A summary of the results and an outlook on possible further investigations is given in the last Chapter.

2 Concepts and theory of turbulence

Turbulence is a phenomenon found in any fluid under motion. On Earth this mainly includes hydrodynamic (HD) fluids, e.g. the atmosphere or water. In the first section of this chapter we introduce the governing equations known for centuries and discuss phenomenologically the groundbreaking work on the statistics of turbulence and its implications by the Russian mathematician Kolmogorov.

In the Earth's core the motion of the electrically conducting fluid metal can be described as a magnetohydrodynamic (MHD) fluid and thus also exhibit turbulence. This description may also be applied to a plasma, which is an ionized gas in the first approximation. Plasmas are scarcely found on Earth, however highly abundant in space, e.g. the constant stream of plasma originating from the Sun called "solar wind". The theoretical description of such MHD fluids and differences in properties of the turbulence compared to HD are introduced in the second section of this chapter.

2.1 Hydrodynamic turbulence

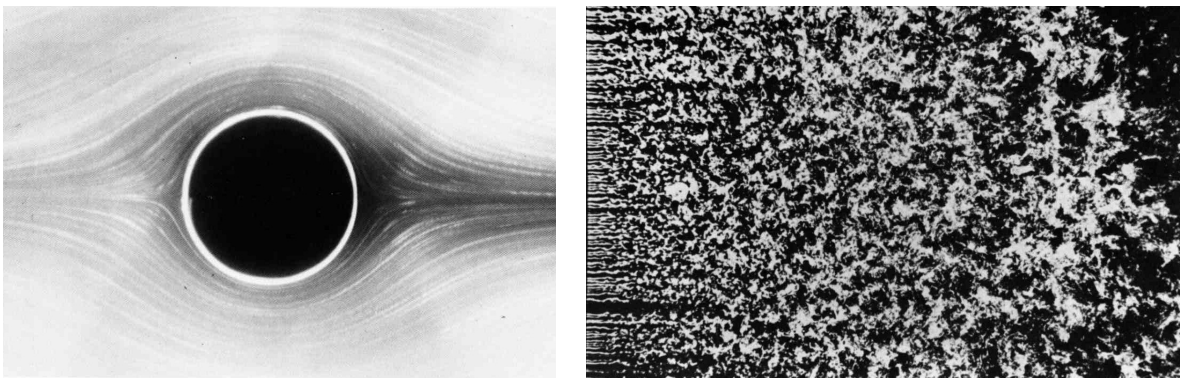
The flow of fluids is described by the Navier-Stokes equations known since Navier (1823), which read in the incompressible formulation

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{F}, \tag{2.1}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{2.2}$$

They consist of the momentum equation (2.1) and the mass conserving continuity equation (2.2), which here reduces to the solenoidality of the velocity $\mathbf{u}$, since the density ρ is constant. The momentum equation includes a time dependent term $\partial_t \mathbf{u}$, a non-linear or convective term $(\mathbf{u} \cdot \nabla)\mathbf{u}$, a pressure gradient term $-\frac{1}{\rho}\nabla p$, a viscous term $\nu \nabla^2 \mathbf{u}$, where ν is the kinematic viscosity, and an additional forcing term $\mathbf{F}$, e.g. gravitational forces $\rho \mathbf{g}$.

A popular parameter to distinguish the type of the flow is introduced to quantify the ratio of dominant terms in the momentum equation. The so-called Reynolds number R is the ratio between the non-linear term $(\mathbf{u} \cdot \nabla)\mathbf{u}$ and the viscous term $\nu \nabla^2 \mathbf{u}$ after dimensional analysis (Reynolds, 1883). Dimensional analysis means that $\mathbf{u}$ is approximated by the characteristic velocity U in the system, e.g. the root mean square (rms) of the amplitudes



(a) **(b)**

Figure 2.1: Laminar flow (a) and turbulent flow (b) for Reynolds numbers $R = 0.16$ and $R > 1500$, respectively (Van Dyke, 1982). The turbulent flow in (b) evolves from left to right down the stream to fully developed (isotropic) turbulence.

of $\mathbf{u}$. The differential operator ∇ is approximated by $1/L$, where L is the characteristic length scale of the system, i.e. the scale on which energy is induced to the system. The Reynolds number is then given by

$$R = \frac{UL}{\nu}. \tag{2.3}$$

If the viscous term is dominant the Reynolds number is low and the flow is found to be laminar. The viscosity then dominates over non-linear convection and any development of chaotic, turbulent motion is ceased. In a laminar flow particle trajectories are traceable, as can be seen in Figure 2.1a. A laminar flow may also be achieved for very large Reynolds numbers, but only for an idealized, perfectly unperturbed flow. In the other case, when the convective non-linear term dominates the system, the Reynolds number is large and the flow is found to be chaotic and/or turbulent. Here, the flow is nondeterministic in time, see Figure 2.1b, so that individual particle trajectories are unpredictable. The critical Reynolds number, which separates laminar from turbulent flow, depends on the system under consideration. It is observed that for very large Reynolds numbers $R \gtrsim 100$ the turbulent motion restores symmetries in a statistical sense and one speaks of fully developed turbulence (Frisch, 1995). A simple example for such a symmetry is the similarity of the distribution function of the velocity fluctuations along different axes in space. This has led theoreticians to develop a probabilistic description of turbulent flow fields (Taylor, 1935, 1938). These statistical symmetries found the basis of the theory by Kolmogorov, introduced in the following.

2.1.1 Kolmogorov theory

In 1941 Andrey Nikolaevich Kolmogorov revolutionized the theory on turbulence by his works on statistical mathematics and symmetries and their application to turbulent flows

(Kolmogorov, 1941a,b,c). Here, we introduce the ideas and the phenomenological consequences following Frisch (1995).

In fully developed turbulence it is found that properties of the flow are spatially invariant in a statistical sense. Furthermore, velocity fluctuations $\delta\mathbf{u}(\mathbf{r}, l) = \mathbf{u}(\mathbf{r}+l) - \mathbf{u}(\mathbf{r})$ at a certain scale l, much smaller than the inertial scale l_0, are spatially homogeneous. Meaning that, in a statistical sense, the fluctuations are invariant under some translation $\mathbf{a}$ in space so that

$$\delta\mathbf{u}(\mathbf{r} + \mathbf{a}, l) = \delta\mathbf{u}(\mathbf{r}, l). \tag{2.4}$$

Another important property that follows from the homogeneity of the fluctuations is the self-similarity of the fluctuations

$$\delta\mathbf{u}(\mathbf{r}, \lambda l) = \lambda^n \delta\mathbf{u}(\mathbf{r}, l), \tag{2.5}$$

with $n, \lambda \in \mathbb{R}$ the scaling exponent and factor, respectively. The third postulation by Kolmogorov is the presence of a mean energy dissipation rate ε. Using these three properties, Kolmogorov derived from the Navier-Stokes equations the exact relation

$$\langle(\delta u(\mathbf{r}, l))^3\rangle = -\frac{4}{5}\varepsilon l, \tag{2.6}$$

which states the ensemble average $\langle\cdot\rangle$ of the third order structure function $(\delta u(\mathbf{r}, l))^3$ is given by the mean energy dissipation rate ε and the scale l in homogeneous isotropic turbulence.

From this exact relation follows phenomenologically the turbulent cascade or Richardson cascade. The energy input rate $\dot{E}$ is equal to the energy dissipation rate ε, implying the existence of a so-called inertial range, where energy is transported from larger scales to smaller scales at exactly that energy input/dissipation rate. If we consider a system, where energy is injected at the inertial scale l_0 at a rate $\dot{E}$ and the energy is dissipated at the scale η_d with a rate ε, the scales $l_0 \gg l \gg \eta_d$ are in the inertial range. In this regime the typical velocity at scale l is estimated as $u_l \sim \sqrt{\langle\delta u^2(l)\rangle}$, the rms of the fluctuations ("$\sim$" denotes "on the order of"). The time scale $\tau_l \sim l/u_l$ associated with these fluctuations is the so-called eddy turnover time after which an "eddy" of this scale has significantly distorted itself. Assuming the rate of energy transfer in the inertial range $\dot{E}_l$ is given by the kinetic energy per eddy turnover time

$$\dot{E}_l \sim \frac{u_l^2}{\tau_l} \sim \frac{u_l^3}{l}, \tag{2.7}$$

and that this energy transfer rate $\dot{E}_l$ is on the order of the dissipation rate ε, we derive that

$$\varepsilon \sim \frac{u_l^3}{l}, \tag{2.8}$$

$$\Leftrightarrow u_l \sim (\varepsilon l)^{1/3}. \tag{2.9}$$

For the power spectral density $E(k)$ per wave number $k \sim l^{-1}$ with $E = \int E(k)dk \approx E(k)k$ follows the well-known power law

$$E(k) \sim Ek^{-1} \sim u_l^2 k^{-1} \sim \varepsilon^{2/3} k^{-5/3}, \tag{2.10}$$

known as the Kolmogorov spectrum. The inertial range reaches down to the Kolmogorov length or dissipation scale

$$\eta_d \sim \left(\frac{\nu^3}{\varepsilon}\right)^{1/4}, \tag{2.11}$$

which can simply be derived by considering the dimensions of $\nu \sim \text{length}^2/\text{time}$ and $\varepsilon \sim \text{length}^2/\text{time}^3$. Similarly the Kolmogorov or dissipation time can be derived as

$$\tau_\eta \sim \left(\frac{\nu}{\varepsilon}\right)^{1/2}. \tag{2.12}$$

When considering scales $l \ll \eta_d$ or times $\tau \ll \tau_\eta$ kinetic effects are non-negligible and a fluid description is invalid.

Turbulence research has mainly focused on the spatial or wave number domain, due to the famous Kolmogorov spectrum. However, as the flow is certainly evolving in time, the temporal aspects must give insights into the flow dynamics and the turbulent decay. Some important time scales and some approaches on the temporal aspects are introduced in the following section.

2.1.2 Characteristic times and frequency spectra in HD turbulence

As already introduced in the previous section, the eddy turnover time τ_l determines the time of energy transfer between eddies. It characterizes on which time scales an eddy of scale l distorts or breaks down into eddies of a smaller scale. In hydrodynamics τ_l is simply equal to the non-linear time

$$\tau_{\text{nl}} \sim \frac{l}{u_l}, \tag{2.13}$$

associated with non-linear distortion. It decreases with scale l. The largest time scale τ_0 of the system, associated with the inertial scale l_0, is often referred to as the large eddy turnover time.

To infer the implication of the non-linear time on the frequency spectrum in a Eulerian rest frame of the fluid (see also Section 4.2.2) Weizsäcker (1948) proposed for the

derivation of a frequency spectrum that u of the dispersion relation $\omega \sim vk$ also scales after Kolmogorov with $v \sim u_l \sim (\varepsilon l)^{1/3}$. From this follows that $\omega \sim \varepsilon^{1/3}k^{2/3}$. If then $E(\omega)d\omega \sim E(k)dk$ and $E(k) \sim k^{-5/3}$, the Eulerian frequency spectrum reads

$$E^{\mathrm{E}}(\omega) \sim \varepsilon\omega^{-2}. \tag{2.14}$$

However, this scaling of the Eulerian frequency spectrum has not been confirmed and a study from Tennekes (1975) proposed a new effect that characterizes the Eulerian frequency spectrum. They assume that the velocity u of $\omega \sim uk$ does not scale after Kolmogorov, but is the rms velocity of the fluctuations at the inertial scale $u' = \sqrt{\langle \delta u^2(l_0) \rangle}$, and so the resulting Eulerian frequency spectrum reads

$$E^{\mathrm{E}}(\omega) \sim \varepsilon^{2/3} v_{\mathrm{rms}}^{2/3} \omega^{-5/3}. \tag{2.15}$$

The derivation is based on the idea known as random sweeping hypothesis, first introduced by Kraichnan (1964). Eddies in the inertial range are said to be dominated by random advection of larger scale fluctuations, namely the rms fluctuations u' at the inertial scale. In simpler words, fluctuations of smaller eddies are "swept" over a fixed point in Eulerian space by larger eddies. The result is a doppler shift and thus spectral broadening, compared to the spectrum derived by Weizsäcker (1948). Several following studies observed Eulerian frequency spectra in support of this hypothesis with a spectral index of $-5/3$ (e.g. Kit et al. 1995). The characteristic time is often referred to as the random sweeping time

$$\tau_{\mathrm{sw}} \sim \frac{l}{u'}. \tag{2.16}$$

In the Lagrangian framework, following the fluid particle/element (detailed definition in Section 4.2.3), different studies have derived a scaling of the frequency spectrum similar to that of Weizsäcker (1948) with

$$E^{\mathrm{L}}(\omega) = \beta_a \varepsilon \omega^{-2}. \tag{2.17}$$

The studies estimate different values for $\beta_a \sim 1$ (Inoue, 1951; Corrsin, 1963; Tennekes and Lumley, 1972; Kaneda, 1993). The calculation of Lagrangian spectra has only been possible since the development of particle tracers in numerical simulations. Since then the spectral index of -2 has been confirmed by several studies (e.g. Lien et al. 1998).

2.2 Magnetohydrodynamic turbulence

Magnetohydrodynamics (MHD) is the flow description of electrically conducting fluids. Such fluids include highly conducting liquid metals which, for example, are found in the Earth's liquid outer core, and plasmas. In addition to the forces present in hydrodynamics the Lorentz force has to be considered, acting on the electrically conducting fluid elements.

It also implies the presence of electric and magnetic fields, the latter playing an important role in the distribution of energy in MHD turbulence. Unlike a mean flow velocity $\mathbf{u}_0$ a mean magnetic field $\mathbf{B}_0$ cannot be removed via Galileo transformation and thus, a preferred direction is introduced to the system. Another consequence of the magnetic field and the Lorentz force is the presence of waves in MHD, introduced in the second part of this section. Their non-linear interactions introduce new mechanisms of energy transfer in turbulence (Iroshnikov, 1964; Kraichnan, 1965). Different models for MHD turbulence based on these properties are presented in the third part of this section. The presence of waves and energy distribution depending on magnetic field direction give rise to new time scales characterizing the flow field and the turbulent decay. They are introduced in the last part of this section. First of all, the governing equations describing an incompressible MHD system are presented.

2.2.1 Incompressible MHD

The equations governing MHD can be derived in a heuristic way, considering which forces act on an electrically conducting fluid. For a plasma the equations are derived by taking the velocity moments of the Vlasov equation, with the moments of the underlying distribution function corresponding to density, velocity and temperature. Since this work is not limited to MHD in a plasma a heuristic derivation is chosen here, following Biskamp (2003) and Baumjohann and Treumann (2012).

In MHD the Lorentz force

$$\mathbf{F}_L = \rho_c(\mathbf{E} + \mathbf{u} \times \mathbf{B}) = \rho_c \mathbf{E} + \mathbf{j} \times \mathbf{B}, \tag{2.18}$$

has to be considered, where ρ_c is the charge density, $\mathbf{E}$ the electric field, $\mathbf{B}$ the magnetic field and $\mathbf{j} = \rho_c \mathbf{u}$ the current density. The momentum equation of incompressible MHD therefore reads

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu \nabla^2 \mathbf{u} + \rho_c \mathbf{E} + \mathbf{j} \times \mathbf{B} + \mathbf{F}. \tag{2.19}$$

Since this equation includes the current density as a new physical quantity an additional equation for the evolution of $\mathbf{j}$ is needed, given by the resistive Ohm's law

$$\mathbf{E} + \mathbf{u} \times \mathbf{B} = \frac{\mathbf{j}}{\sigma_0}, \tag{2.20}$$

which is a simplification of the generalized Ohm's law under the assumptions that the spatial and temporal scales are large enough. The conductivity is given by σ_0. When inserting this resistive Ohm's law into Faraday's law of induction

$$\partial_t \mathbf{B} = -\nabla \times \mathbf{E} \tag{2.21}$$

$$= \nabla \times \left(\mathbf{u} \times \mathbf{B} - \frac{\mathbf{j}}{\sigma_0} \right) \tag{2.22}$$

$$= \nabla \times (\mathbf{u} \times \mathbf{B}) - \frac{1}{\sigma_0} \nabla \times \mathbf{j}, \tag{2.23}$$

and making use of Ampere's law $\mu_0 \mathbf{j} = \nabla \times \mathbf{B}$, where μ_0 is the magnetic permeability, the induction equation for resistive MHD is derived. Due to a divergence free magnetic field, $\nabla \cdot \mathbf{B} = 0$, this induction equation reads

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}, \tag{2.24}$$

with $\eta = (\sigma_0 \mu_0)^{-1}$ the magnetic diffusivity. The continuity equation $\nabla \cdot \mathbf{u} = 0$ remains the same as in incompressible HD, so the full set of incompressible MHD equations read

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} + \rho_c \mathbf{E} + \mathbf{j} \times \mathbf{B} + \mathbf{F}, \tag{2.25}$$

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}, \tag{2.26}$$

$$\nabla \cdot \mathbf{u} = 0, \tag{2.27}$$

$$\nabla \cdot \mathbf{B} = 0. \tag{2.28}$$

For a plasma this approach is valid for scales and times larger than the characteristic scales and times of the plasma. The length scales must be larger than the ion gyro radius

$$r_i = \frac{m v_\perp}{q|\mathbf{B}|}, \tag{2.29}$$

where m is the ion mass, $v_\perp$ the speed of the ion perpendicular to $\mathbf{B}$ and q the elementary charge. The time scales should be much larger than the period of the ion gyro frequency

$$\omega_i = \frac{q|\mathbf{B}|}{m}. \tag{2.30}$$

Otherwise a kinetic approach is needed to account for single particle motions, drift effects and kinetic waves (Baumjohann and Treumann, 2012).

Magnetic Reynolds number

In addition to the classical Reynolds number (2.3) the MHD flow can also be characterized by the magnetic Reynolds number R_m. It is given by the ratio of the convective term

$\nabla \times (\mathbf{u} \times \mathbf{B}) \sim UB/L$ and the diffusive term $\eta \nabla^2 \mathbf{B} \sim B/(L^2 \mu_0 \sigma_0)$ of the induction equation, so that

$$R_m = \mu_0 \sigma_0 L U. \tag{2.31}$$

If the magnetic Reynolds number is small $R_m \sim 1$ the diffusive term dominates over the convective term and the magnetic field is not influenced by the flow. For a large magnetic Reynolds number, $R_m \gg 1$, the convective term of the induction equation dominates over the diffusive term. The magnetic field can be seen as frozen into the flow field (Baumjohann and Treumann, 2012).

Frozen-in field theorem

A large magnetic Reynolds number $R_m \gg 1$ is found for very large conductivities σ_0 or very large length scales, so that Ohm's law reduces to

$$\mathbf{E} = -\mathbf{u} \times \mathbf{B}, \tag{2.32}$$

and the induction equation reads

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{u} \times \mathbf{B}) \,. \tag{2.33}$$

This is called non-resistive or ideal MHD and one of the consequences is the so-called frozen-in field theorem (Alfvén, 1943; Baumjohann and Treumann, 2012). It can easily be shown that the magnetic flux through any surface

$$\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} = 0 \tag{2.34}$$

vanishes in this approximation (Biskamp, 2003). The magnetic flux through any surface moving in the plasma is then conserved and the magnetic field can be regarded as "frozen" into the flow. Another consequence is the conservation of topology, which is known to be broken in magnetic reconnection (Biskamp, 2003). The actual coupling between magnetic field and velocity fluctuations is unclear, but in a regime where the frozen-in field theorem holds a high correlation between the fields is expected.

Elsässer variables

By introducing the so-called Elsässer variables $\mathbf{z}^{\pm} = \mathbf{u} \pm \mathbf{b}$, where $\mathbf{b} = \mathbf{B}/\sqrt{\mu_0 \rho}$ is the magnetic field in velocity units, the incompressible MHD equations (2.25)-(2.28) can be reduced to

$$\partial_t \mathbf{z}^{\pm} + (\mathbf{z}^{\mp} \cdot \nabla)\mathbf{z}^{\pm} = -\frac{1}{\rho} \nabla p + \nabla^2 \left(d_{\pm} \mathbf{z}^{+} + d_{\mp} \mathbf{z}^{-} \right), \tag{2.35}$$

$$\nabla \cdot \mathbf{z}^{\pm} = 0, \tag{2.36}$$

where $d_\pm = (\nu \pm \eta)/2$ (Elsässer, 1950). This has implications on the physical properties of an MHD fluid, as the analogy to the hydrodynamic equations (2.1)-(2.2) is clear and therefore we also expect turbulence to occur. However, the convective non-linear term $(\mathbf{z}^\mp \cdot \nabla)\mathbf{z}^\pm$ shows opposing signs of the Elsässer variables. Non-linear interactions thus only occur by fluctuations in opposing directions. In the following section we will introduce waves in MHD and find that these opposing fluctuations can be seen as counter-propagating Alfvén waves.

2.2.2 Waves in MHD

In contrast to hydrodynamics, where all particle interactions are non-linear and collisional, waves are present in MHD. By linearizing the compressible MHD equations through a perturbation approach, so that $\rho = \rho_0 + \delta\rho$, $\mathbf{u} = \delta\mathbf{u}$, $p = \delta p$, $\mathbf{B} = \mathbf{B}_0 + \delta\mathbf{B}$ and $\mathbf{E} = \delta\mathbf{E}$, a linear system of equations is given with dispersion relations as the eigenmodes of this system. These dispersion relations correspond to the Alfvén waves and the magnetosonic waves, the latter being only present in compressible MHD (Alfvén, 1942; Baumjohann and Treumann, 2012). Besides these wave modes numerous other dispersion relations, corresponding to different plasma waves, can be derived (Baumjohann and Treumann, 2012; Stix, 1992). Since this thesis is limited to incompressible MHD we only introduce the present Alfvén mode. When linearizing the system the linear dispersion relation $\omega = \pm u_A k_\parallel$ is derived, where $u_A = |\mathbf{B}_0|/\sqrt{\mu_0\rho_0}$ is defined as the Alfvén velocity. This relation is decoupled from all directions not parallel to $\mathbf{B}_0$. The corresponding transversal waves are called Alfvén waves, which are non-dispersive ($v_{\text{group}} = v_{\text{phase}} = u_A$), propagate parallel or antiparallel to $\mathbf{B}_0$ and whose amplitudes are perpendicular to $\mathbf{B}_0$. The restoring force is the magnetic tension, occurring when bending magnetic field lines.

2.2.3 Phenomenological models in MHD turbulence

Additional dynamics in the form of waves and a present magnetic field are found in MHD and more drivers for turbulent decay are expected. As a magnetic field is not removable by Galileo transformation compared to a mean flow velocity, a preferred direction is introduced and anisotropy is also expected in the statistical symmetries of a fully developed turbulent system. New characteristic times in MHD are found and presented here in relation to MHD turbulence models.

Irochnikov-Kraichnan theory

Iroshnikov (1964) and Kraichnan (1965) (IK) suggested a new scaling for the inertial range of the turbulent cascade, based on Alfvénic interactions. As can be seen from the MHD equations in the Elsässer formulation, only counter propagating Alfvén waves interact non-linearly. IK propose that the time of counter propagating Alfvén waves,

which they define as the Alfvén time $\tau_{\mathrm{A}} = l/u_A$, is much smaller than the eddy distortion time $\tau_{z,l}^{\pm} = l/\delta z_l^{\mp}$ and the relative change of amplitude $\Delta\delta z_l^{\pm}/\delta z_l^{\pm}$ is small, so

$$\frac{\Delta\delta z_l^{\pm}}{\delta z_l^{\pm}} = \frac{\tau_{\mathrm{A}}}{\tau_{z,l}^{\pm}} \ll 1, \tag{2.37}$$

where $\delta z_l^{\pm}$ are the fluctuation of the Elsässer fields of scale l. IK further suggest that the number of interactions is dominated by isotropic diffusion and to achieve a relative change of the amplitudes on the order unity $N \sim (\delta z_l^{\pm}/\Delta\delta z_l^{\pm})^2$ interactions are needed. The eddy turnover time thus equals N times the time of counter propagating wave packets

$$\tau_l^{\pm} \sim N\tau_{\mathrm{A}} \sim \frac{(\tau_{z,l}^{\pm})^2}{\tau_{\mathrm{A}}}. \tag{2.38}$$

Since this turnover time is longer than the non-linear time $\tau_{\mathrm{nl}} = l/u_l$ the amplitudes of decreasing scales l increase. The corresponding spectrum reads

$$E(k) \sim (\varepsilon u_A)^{1/2} k^{-3/2}, \tag{2.39}$$

has a shallower spectral index $-3/2$ compared to the $-5/3$ in HD. The biggest scepticism with the IK theory is the assumption of isotropic diffusion of Alfvén waves. As the magnetic field introduces a preferred direction, along which Alfvén waves propagate, the assumption of isotropy seems implausible (Montgomery and Turner, 1981). Therefore several theories have been developed considering directional dependencies (Matthaeus et al., 1990; Goldreich and Sridhar, 1995; Bieber et al., 1996; Boldyrev, 2006; Oughton et al., 2015).

Slab + 2D turbulence

The first anisotropic model was introduced by Matthaeus et al. (1990). They assume that the fluctuation energy is not isotropically distributed, as the IK theory suggest, but into a 'slab' component $k_{\parallel}$ parallel and a '2D' component $k_{\perp}$ perpendicular to a mean magnetic field $\mathbf{B}_0$. Weak Alfvénic interactions are proposed to be parallel to the mean magnetic field and non-linear interactions are purely perpendicular. The model has led to multiple theoretical predictions, being supported by anisotropic spectra from spacecraft observations (Bieber et al., 1996; Saur and Bieber, 1999). However, the assumption of energy being purely distributed along and perpendicular to a mean magnetic field seems unrealistic compared to a distribution in the whole $\mathbf{k}$-space.

Critical balance

Goldreich and Sridhar (1995) proposed a new model known as 'critically balanced' turbulence. Their idea is a balance between the non-linear time $\tau_{\rm nl}$ and the Alfvén time $\tau_{\rm A}$ so that

$$\tau_{\rm nl} \sim \tau_{\rm A} \Leftrightarrow k_\perp u_k \sim k_\parallel u_A, \tag{2.40}$$

where $u_k = u_l$. The energy is then distributed axisymmetrically in the $\mathbf{k}$-space

$$E(k_\parallel, k_\perp) \sim \frac{u_A^2}{k_\perp^{10/3} l_0^{1/3}} f\left(\frac{k_\parallel l_0^{1/3}}{k_\perp^{2/3}}\right), \tag{2.41}$$

depending on the inertial scale l_0. It can be shown that under the critical balanced assumption (2.40) the parallel and perpendicular energy spectra are obtained by

$$E(k_\parallel) \sim \varepsilon u_A^{-1} k_\parallel^{-2} \quad \text{and} \quad E(k_\perp) \sim \varepsilon^{2/3} k_\perp^{-5/3}. \tag{2.42}$$

Several studies confirmed the anisotropic scalings given in equation (2.42), leading to a large acceptance of the theory of critical balance theory in the community (Horbury et al., 2008; Podesta, 2009; Wicks et al., 2010).

2.2.4 Characteristic times in MHD turbulence

In MHD characteristic times have been introduced in addition to those already present in HD. Besides the non-linear interaction time $\tau_{\rm nl}$ and the random sweeping time $\tau_{\rm sw}$ the Alfvén time $\tau_{\rm A}$ and a new eddy transfer time τ_l have been introduced. Servidio et al. (2011) introduce the non-linear time to be

$$\tau_{\rm nl} \sim (k\sqrt{kE^\pm(k)})^{-1}, \tag{2.43}$$

with $E^\pm(k)$ the Elsässer energy density spectrum. For the Kolmogorov scaling this can be rewritten to

$$\tau_{\rm nl}(k) \sim \left(C_\mp^{1/2} \varepsilon_\mp^{2/3} \varepsilon_\pm^{-1/3} k^{2/3}\right)^{-1}, \tag{2.44}$$

where $\varepsilon_\pm = -\frac{d}{dt}\langle \frac{1}{2}\left(\mathbf{z}^\pm\right)^2\rangle = \langle \nu \left(\nabla \times \mathbf{z}^\pm\right)^2\rangle$ are the dissipation rates. They also define the random sweeping time

$$\tau_{\rm sw}^\pm(k) \sim (k(z^\mp)')^{-1}, \tag{2.45}$$

the Alfvén time

$$\tau_{\rm A}(k) \sim (kb')^{-1}, \tag{2.46}$$

and the viscous dissipation time

$$\tau_d(k) \sim \left(\nu k^2\right)^{-1}. \tag{2.47}$$

All these times are defined under the assumption of isotropy, meaning $\mathbf{B}_0 = \mathbf{0}$. The isotropic assumption should be viewed with scepticism, especially for the Alfvén time, which in most cases is defined parallel to a mean magnetic field. Also, the definitions of the characteristic times are oriented on the Elsässer fields, not the velocity field, which Lugones et al. (2016) use to define the characteristic times.

Besides the implications of the Alfvén time for wave number spectra the literature on the temporal domain is scarce and has only been under considerable interest in the last few years. Early work has been done by Pouquet (1993) who picked up on the frequency spectrum after Weizsäcker (1948), but independently derived a Eulerian frequency spectrum for MHD. Her approach was to set the eddy transfer time to be $\tau_l \sim \tau_{\mathrm{nl}}^2/\tau_{\mathrm{A}}$, following the IK phenomenology. The energy dissipation rate ε is then given by

$$\varepsilon \sim \frac{\omega E(\omega)}{\tau_l} \sim \frac{\omega E(\omega)}{\tau_{\mathrm{nl}}^2/\tau_{\mathrm{A}}}. \tag{2.48}$$

Using the non-linear transfer time $\tau_{\mathrm{nl}} \sim \omega_{\mathrm{nl}}^{-1} \sim \omega^{-1}$, the Alfvén time $\tau_A = (ku_A)^{-1}$, with u_A the Alfvén velocity, and $k \sim \omega/v \sim (E(\omega)\omega^{-1})^{-1/2}$ it follows

$$\varepsilon \sim \frac{\omega^3 E(\omega)}{u_A}\left(\frac{E(\omega)}{\omega}\right)^{1/2} \tag{2.49}$$

$$\Leftrightarrow E^{\mathrm{E}}(\omega) \sim (\varepsilon u_A)^{2/3}\omega^{-5/3}. \tag{2.50}$$

This is very similar to the Eulerian frequency spectrum of hydrodynamics derived by Tennekes (1975), only with the Alfvén velocity instead of the rms velocity. Physically, the fluctuations are then swept over a fixed point by the large scale Alfvén waves instead of large scale motions.

In the Lagrangian framework Busse et al. (2010) derived two possible spectra considering strong or weak non-linear interaction. Their definition of strong and weak interaction is based on correlation times, whereas in the literature often $\delta\mathbf{b} \ll \mathbf{B}_0$ is considered as weak interaction and $\delta\mathbf{b} \sim \mathbf{B}_0$ as strong interaction. Busse et al. (2010) state that weak interactions are present, if $\tau_D \ll \tau_l$, where $\tau_D = \tau_D(k)$ is the decorrelation time and τ_l is the eddy turnover time. Strong interaction is said to be present, if $\tau_D \sim \tau_l$. In contrast to the Eulerian frequency spectra, these differences in characteristic times play an important role in the Lagrangian spectrum. Based on strong interaction they derive a spectrum

$$E^{\mathrm{L}}(\omega) \sim \varepsilon\omega^{-2}, \tag{2.51}$$

which is expected with theories from Goldreich and Sridhar (1995) or Boldyrev (2006), where $\tau_D \sim \tau_{\mathrm{A}} \sim \tau_{\mathrm{nl}} \sim \tau_l$. Based on weak interactions, following the IK theory with $\tau_D \sim \tau_{\mathrm{nl}} \ll \tau_{\mathrm{nl}}^2/\tau_{\mathrm{A}} \sim \tau_l$, results in

$$E^{\mathrm{L}}(\omega) \sim \varepsilon^{1/2}u_A\omega^{-3/2}, \tag{2.52}$$

where u_A is again the Alfvén velocity (Busse et al., 2010).

3 MHD simulations

To study flow field dynamics in an MHD fluid, two options are available. First, by analyzing experimental or observational data, obtained from laboratory measurements or spacecraft, and second, by numerical simulations. Laboratory MHD plasmas are usually very difficult to measure, due to the temperature, pressure and strong magnetic fields needed to confine such a plasma (e.g. Yamada et al. 1997). Similar to the data obtained by spacecraft, the measurements are point probes, making it impossible to analyze any Lagrangian dynamics. Spacecraft data is also inevitably exposed to measure time series as a mixture of spatial and temporal dynamics, in the spacecraft rest frame (Matthaeus et al., 2016). To achieve a full 3D spatial and temporal resolution of the system direct numerical simulations (DNS) are needed. These have only become feasible in the last two to three decades with increasingly large computational power available. They allow the analysis of purely spatial and purely temporal dynamics of the flow as well as Lagrangian properties. The simulations used here were carried out by the John's Hopkins University and are made publicly available in a database (Li et al., 2008). In this chapter we introduce the database, the simulations analyzed in this thesis and the forcing applied in the MHD simulation.

3.1 John's Hopkins Turbulence Database

To analyze temporal dynamics and trace particles the full temporal resolution is needed. Only recently this very hardware consuming task has been tackled by the John's Hopkins Turbulence Database (JHTDB). The aim is to make the whole 4D simulation results available for analysis. It was realized by sophisticated database storage and access via interpolation schemes (Li et al., 2008). The idea is that a minimum amount of data needed for the reconstruction is stored on the server side. The database is accessed through a Web services interface C-library ported to the Julia Programming language (Bezanson et al., 2017). On the server side the queried data is calculated using the interpolation schemes and then send back to the user. In this way the storage of the large amount of data is possible.

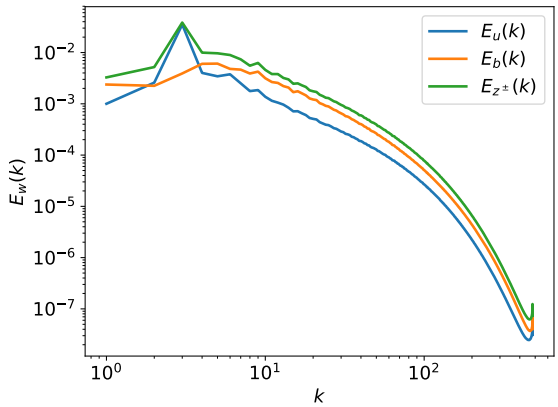


Figure 3.1: Power spectral densities $E_w(k)$ for w the velocity (blue), magnetic field (orange) and the positive Elsässer variable (green) in the MHD simulation.

3.2 Forced isotropic incompressible MHD simulation

The incompressible MHD equations in the Elsässer formulation

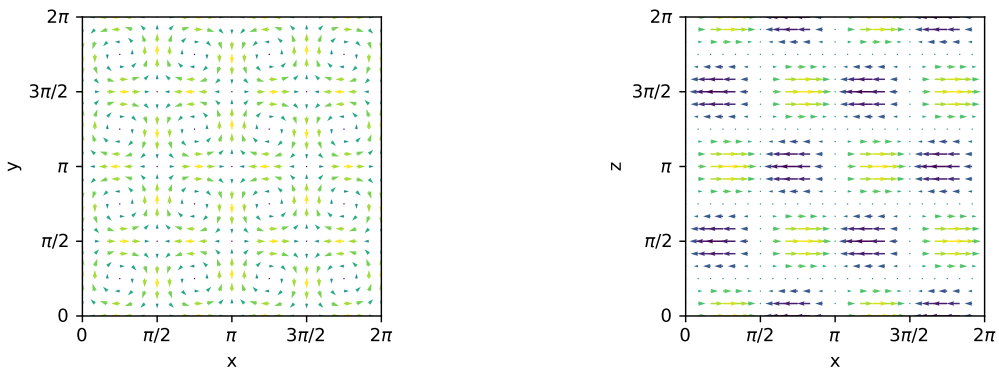
$$\partial \mathbf{z}^{\pm} + (\mathbf{z}^{\mp} \cdot \nabla)\mathbf{z}^{\pm} = -\nabla p + \nu \nabla^2 \mathbf{z}^{\pm} + \mathbf{F}^{\mathbf{u}}, \tag{3.1}$$

$$\nabla \cdot \mathbf{z}^{\pm} = 0, \tag{3.2}$$

with $\rho \equiv 1$ are solved in a $2\pi \times 2\pi \times 2\pi$ box with 1024^3 grid points and periodic boundary conditions using a pseudo-spectral solver (Aluie, 2009; Eyink et al., 2013). The pseudo-spectral solver uses the convolution theorem and efficiently calculates spatial derivatives in Fourier space. To account for aliasing effects introduced by the pseudo-spectral solver a $2\sqrt{2/3}$ truncation is applied (Patterson and Orszag, 1971). The magnetic Prandtl number $Pr = \nu/\eta$ is set to unity with $\nu = \eta = 1.1 \times 10^{-4}$. The forcing $\mathbf{F}^{\mathbf{u}}$ is applied to the flow-field only and will be discussed in detail in the following section. The characteristic quantities of the simulation are given in Table 3.1. Once the simulation has reached a

Table 3.1: Characteristic quantities of the analyzed MHD and HD simulation. The parameters can be calculated for each field $\mathbf{w} \in \{\mathbf{u}, \mathbf{b}\}$. Given are the dissipation rates $\varepsilon_w = 2\nu \int k^2 E_w(k) dk$, the rms values $w' = \langle |\mathbf{w}|^2/3 \rangle^{1/2}$, the integral scales $l_{0,w} = \pi/(2w'^2) \int k^{-1} E_w(k) dk$, the large eddy turnover times $\tau_w = l_{0,w}/w'$, the Kolmogorov length scales $\eta_w = (\nu^3/\varepsilon_w)^{1/4}$, the Kolmogorov time scales, $\tau_{\eta,w} = (\nu/\varepsilon_w)^{1/2}$, the Taylor micro scale $\lambda_w = (15\nu/\varepsilon_w)^{1/2} w'$, the Taylor-scale Reynolds number $R_{\lambda,w} = w'\lambda_w/\nu$ and the kinetic and magnetic Reynolds numbers with $R_u = u' l_{0,u}/\nu$ and $R_b = \mu_0 \sigma b' l_{0,b}$, respectively.

Sim.	$\mathbf{w}$	ε_w	w'	$l_{0,w}$	τ_w	η_w	$\tau_{\eta,w}$	λ_w	$R_{\lambda,w}$	R
MHD	$\mathbf{u}$	1.1×10^{-2}	0.23	0.56	2.43	3.3×10^{-3}	0.1	8.9×10^{-2}	186	764
	$\mathbf{b}$	2.2×10^{-2}	0.24	0.35	1.46	2.8×10^{-3}	0.07	6.6×10^{-2}	144	1170
HD	$\mathbf{u}$	1.0×10^{-1}	0.69	1.36	1.99	2.8×10^{-3}	0.04	0.113	418	5058



(a) $z = \pi/2$ **(b)** $y = \pi$

Figure 3.2: Visualization of the Taylor-Green forcing applied in the MHD simulation, where (a) shows a cut through the xy-plane and (b) a cut through the xz-plane.

statistically stationary state the results are stored every 10th time integration, so that $\delta t = 2.5 \times 10^{-3}$, for 1024 times until $t_{\max} = 2.56$, resulting in 1024^4 values for the velocity $\mathbf{u}(\mathbf{x}, t)$, the magnetic field $\mathbf{b}(\mathbf{x}, t)$ and the pressure $p(\mathbf{x}, t)$. The analysis of the full 4D space-time domain, necessary for temporal correlations and Lagrangian dispersion in the flow field, is now possible. The wave number energy density spectrum of the velocity, the magnetic field and the positive Elsässer variable are shown in Figure 3.1. A turbulent cascade can be identified, with an energy injection region up to around $k = 10$, a power-law region associated with an inertial range for $10 < k < 50$ and a dissipative regime for $k > 50$. The flow can thus be seen as fully developed turbulence.

3.3 Forcing in the simulation

During the whole time of the simulation a Taylor-Green forcing is applied to the momentum equation (Taylor and Green, 1937). It is only applied to the flow field in the xy-plane so that

$$\mathbf{F}^{\mathbf{u}} = f_0 \left(\sin(k_f x) \cos(k_f y) \cos(k_f z) \mathbf{e}_x - \cos(k_f x) \sin(k_f y) \cos(k_f z) \mathbf{e}_y \right), \tag{3.3}$$

where $f_0 = 1/4$ and $k_f = 2$. The shape of this forcing can be seen as twisted cylinders parallel to the z-axis, as visualized in Figure 3.2. The forcing is maximal in the planes where $z = n\pi/2$ with $n = 0, ..., 4$ and vanishes for $z = n\pi/4$ with $n = 1, 3, 5, 7$. The pattern emerging is π periodic in x, y and z direction. The forcing is constant in time and present during the whole simulation. The injection of energy can clearly be seen in the energy density spectra of the velocity and the Elsässer variable at the energy injection scale $k = 2$ (see Figure 3.1 blue and green lines). The energy injection is not visible in the magnetic field, which is discussed in more detail in Section 5.1.

3.4 Forced isotropic HD simulation

In addition to the MHD simulation we consider an HD simulation as a reference for observed effects in the analysis of the MHD simulation. Similar to the MHD simulation the domain is a $2\pi \times 2\pi \times 2\pi$ periodic box with 1024^3 grid points. The time is stored every $\delta t = 2 \times 10^{-3}$ for approximately 5 large eddy turnover times. Characteristic quantities of the forced isotropic HD simulation, which is also part of the JHTDB database, are given in Table 3.1. Note, that the forcing in the HD simulation is not Taylor-Green forcing, but randomly at $k \leq 2$.

4 Statistical analysis of turbulence

As introduced in Chapter 2 symmetries in a fully developed turbulent flow are only found for the statistical moments of the flow fields. Therefore we want to introduce the assumptions necessary for a statistical analysis and common as well as new tools to study flow dynamics. First, the ergodic theorem is presented, which is needed to infer ensemble averages from finite samples. Correlation functions have been widely used in turbulence analysis, but scarcely with focus on scale-dependence and the temporal domain. They are introduced in the second section. For the analysis of Lagrangian particle dispersion new approaches are introduced in the last section of this chapter.

4.1 Ergodic theorem

Measurements or numerical simulations are only single finite samples of the flow field and do not represent the full ensemble. That is why the ergodic theorem needs to be valid to infer ensemble statistics from observations or simulations. The ergodic theorem states that for any sample $v_i(t)$ of a random process $v(t)$ the ensemble average $\langle v \rangle$ is given by

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T v_i(t)dt = \langle v \rangle, \tag{4.1}$$

if the sample size T is large enough (Birkhoff, 1931; Frisch, 1995). The random process v is than referred to as being ergodic.

This assumption allows us to infer turbulent flow properties from statistical tools when applying them to finite data. It is well accepted that this assumption is valid for fully developed turbulence and has been used in all numerical and observational studies (Galanti and Tsinober, 2004).

4.2 Correlation functions

Correlation functions are one of the oldest tools in statistical analysis. They are used to quantify the dependence between two or more random processes. They can be defined in various ways. First of all, we can distinguish between the Eulerian and the Lagrangian framework. These two viewpoints, their physical interpretation and the definitions of the

two-point space-time correlation functions are presented in this chapter. Both Lagrangian and Eulerian correlation functions are not limited to two-point correlations, but any other correlation can be used, e.g. n-point correlations (Batchelor, 1982). Higher point correlations may be used, for example, for improved robustness or the isolation of specific events (e.g. waves). However, here, we focus on the two-point correlation functions, as they are closely related to the energy density spectrum and have an easy physical interpretation (Batchelor, 1982).

The two-point autocovariance function of a generic one-dimensional stationary random process $v(x)$ is defined as

$$C(\xi) = \langle v(x)v(x+\xi)\rangle_x, \tag{4.2}$$

where x can be a spatial, temporal or any other physical domain, ξ is a lag in x and $\langle\cdot\rangle_x$ denotes the average over x. When normalizing the autocovariance function the more commonly used autocorrelation function

$$R(\xi) = C(\xi)/C(0) = \langle v(x)v(x+\xi)\rangle_x/\langle v^2\rangle_x, \tag{4.3}$$

is obtained (Buttkus, 2000). This function $R(\xi)$ gives insights into the self-similarity of the random process v, as it is compared to itself shifted by a lag ξ. For $\xi = 0$ the correlation function is maximal and equal to 1, which we shall refer to as full correlation. If $R(\xi) = -1$ the process fully anticorrelates, which, for example, would be the case for a sin-function with lag $\xi = \pi$, where the phase is exactly opposite.

4.2.1 Wiener-Khintchine theorem

The two-point autocovariance function $C(\xi)$ of a random process $v(x)$ is also related to its energy density spectrum $E_v(\lambda)$ via the Wiener-Khintchine theorem

$$C(\xi) = \int E_v(\lambda)e^{i\lambda\xi}d\lambda, \tag{4.4}$$

introduced by Wiener (1930) and Khintchine (1934). For ξ being a spatial or temporal lag λ is the corresponding wave number or frequency. As this relates the correlation function to the energy density spectrum, which has been shown to play an important role in turbulence theory and the physical implications of turbulence (see Chapter 2), we also expect physical insights from the correlation functions.

4.2.2 Eulerian frame

In the Eulerian framework, spatial and temporal fluctuations at fixed points in an inertial frame are observed. The Eulerian frame can be either the frame through which the fluid under consideration moves with a mean velocity $\mathbf{u}_0$, e.g. a wind tunnel, or the frame with $\mathbf{u}_0 = \mathbf{0}$, as usually used in numerical simulations. The two-point space-time

crosscovariance function of the random processes v and w in the Eulerian frame is defined as

$$C^{\mathrm{E}}_{vw}(\mathbf{r},\tau) = \langle v(\mathbf{x},t)w(\mathbf{x}+\mathbf{r},t+\tau)\rangle, \tag{4.5}$$

where $\langle\cdot\rangle$ denotes the ensemble average, which is here and in the following, unless denoted differently, equal to the average $\langle\cdot\rangle_{\mathbf{x},t}$ over all sample times t and positions $\mathbf{x}$ according to the ergodic theorem. The correlation function is essentially a convolution of the fluctuations of v at point $\mathbf{x}$ and time t with the fluctuations of w at point $\mathbf{x}+\mathbf{r}$ and time $t+\tau$. Both spatial lag $\mathbf{r}$ and temporal lag τ can be varied independently. When taking the covariance function $C^{\mathrm{E}}_{vv}(\mathbf{r},\tau)$ of a process v with itself one speaks of the autocovariance function. The crosscovariance function is easily extended to full vector processes $\mathbf{v}$ and $\mathbf{w}$ by

$$C^{\mathrm{E}}_{\mathbf{vw}}(\mathbf{r},\tau) = \langle \mathbf{v}(\mathbf{x},t)\cdot\mathbf{w}(\mathbf{x}+\mathbf{r},t+\tau)\rangle = \sum_{i=x,y,z} C^{\mathrm{E}}_{v_i w_i}(\mathbf{r},\tau). \tag{4.6}$$

The relation to the Eulerian wavevector and frequency spectra are given by

$$C^{\mathrm{E}}_{\mathbf{vv}}(\mathbf{r},\tau=0) = \int E^{\mathrm{E}}_{\mathbf{v}}(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{r}}d\mathbf{k}, \tag{4.7}$$

$$C^{\mathrm{E}}_{\mathbf{vv}}(\mathbf{r}=\mathbf{0},\tau) = \int E^{\mathrm{E}}_{v}(\omega)e^{i\omega\tau}d\omega, \tag{4.8}$$

following equation (4.4).

Often, the normalized covariance function, the correlation function

$$R^{\mathrm{E}}_{\mathbf{vw}}(\mathbf{r},\tau) = \frac{\langle \mathbf{v}(\mathbf{x},t)\cdot\mathbf{w}(\mathbf{x}+\mathbf{r},t+\tau)\rangle}{(\langle\mathbf{v}^2\rangle\langle\mathbf{w}^2\rangle)^{1/2}}, \tag{4.9}$$

is shown, where the autocorrelation $R^{\mathrm{E}}_{\mathbf{vv}}(\mathbf{0},0) = 1$. For the autocorrelation (and autocovariance) it can easily be shown that $R^{\mathrm{E}}_{\mathbf{vv}}(\mathbf{r},\tau) \leq R^{\mathrm{E}}_{\mathbf{vv}}(\mathbf{0},0)$ using the Schwartzian inequality (Batchelor, 1982).

When considering finite and discrete time series the convolution can be calculated in different ways, depending on the boundary conditions. For a periodic box $2\pi\times2\pi\times2\pi$ the periodic boundary condition infers that

$$R^{E}_{vw}(\mathbf{r},\tau=0) = R^{E}_{vw}(\mathbf{r}+\mathbf{a},\tau=0), \tag{4.10}$$

for $\mathbf{a}$ in any combination of $2\pi\{\mathbf{e}_x,\mathbf{e}_y,\mathbf{e}_z\}$. For the periodic box of the spatial domain in the simulation the convolution can be calculated periodically using the Fourier transform. This does not apply to the time domain, where the convolution needs to be calculated using an aperiodic boundary. Usually the time series is padded with zeros, so that

$$R^{\mathrm{E}}_{vw}(\mathbf{r},\tau) \xrightarrow{\tau\to T} 0. \tag{4.11}$$

For the analysis of space-time correlations we implement the correlation functions with mixed boundary conditions, periodic for the spatial domain and aperiodic for the temporal domain.

Scale-dependent Eulerian correlation function

To isolate the importance of temporal decorrelation effects scale-dependently the scale-dependent, or filtered, Eulerian correlation function has been introduced to turbulence research (Heisenberg, 1948; Comte-Bellot and Corrsin, 1971; Orszag and Patterson, 1972; Sanada and Shanmugasundaram, 1992; Servidio et al., 2011; Lugones et al., 2016). When Fourier-transforming the space-time autocovariance function (4.6) in the spatial domain the time-lagged wave-vector spectrum

$$S_w^E(\mathbf{k},\tau) = E_w^E(\mathbf{k})\Gamma_w(\mathbf{k},\tau), \tag{4.12}$$

is obtained. It can be factorized into the wave-vector spectrum $E_w^E(\mathbf{k})$ and the scale-dependent correlation function $\Gamma_w(\mathbf{k},\tau)$ (Orszag and Patterson, 1972; Servidio et al., 2011). In practice this scale-dependent autocorrelation function can be calculated by

$$\Gamma_w(\mathbf{k},\tau) = Q_w(\mathbf{k},\tau)/E_w^E(\mathbf{k}), \tag{4.13}$$

with

$$Q_w(\mathbf{k},\tau) = \langle \tilde{\mathbf{w}}(\mathbf{k},t)\tilde{\mathbf{w}}^*(\mathbf{k},t+\tau)\rangle_t, \tag{4.14}$$

with $\tilde{\mathbf{w}}$ being the Fourier transform in the spatial domain of the vector field $\mathbf{w}$ and $\tilde{\mathbf{w}}^*$ denoting its complex conjugate (Sanada and Shanmugasundaram, 1992; Servidio et al., 2011).

The scale-dependency introduced to the correlation function allows a better comparison of decorrelation times and characteristic times in the turbulent system, which are also functions of k.

Application in HD and MHD turbulence

Spatial correlations have been studied since the onset of the statistical analysis of turbulence (Taylor, 1935, 1938; Heisenberg, 1948). The first applications of Eulerian space-time correlation functions to HD turbulence also reach far back in time (Favre et al., 1961; He et al., 2017). The focus has been on the validation of Taylor's hypothesis in laboratory turbulence. Taylor (1938) proposed that if the velocity fluctuations $\delta u(t)$ are small compared to the mean velocity u_0 fluctuations at a fixed point in Eulerian space can be regarded as spatial fluctuations $\delta u(r)$:

$$\delta u(t) \to \delta u\left(\frac{r}{u_0}\right) \qquad \text{, if } \frac{\delta u(t)}{u_0} \to 0. \tag{4.15}$$

This approximation, known as Taylor's hypothesis or Taylor's frozen flow approximation, essentially disregards temporal changes. Clearly, this is a good approximation, if u_0 is large enough. The frequency spectrum $E(\omega)$ can then be translated to the wave number spectrum $E(k)$. For the Eulerian space-time correlation function, the Taylor hypothesis implies

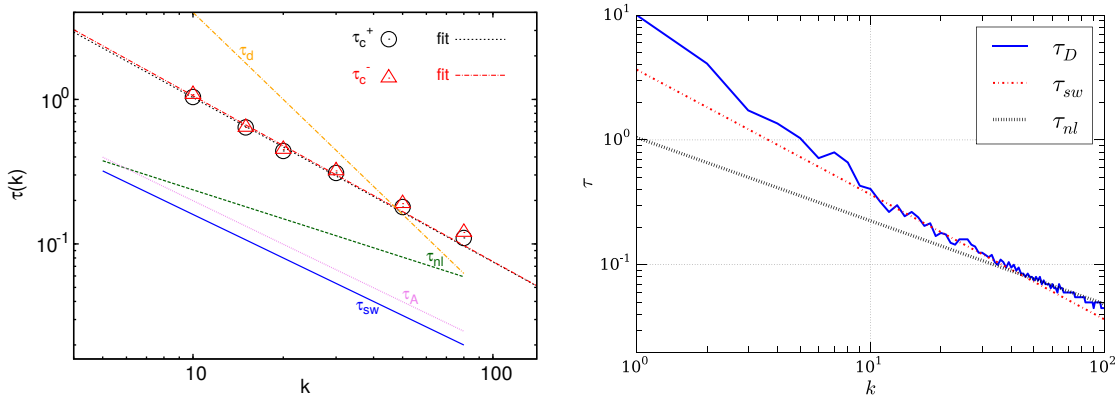
$$R^{\mathrm{E}}_{uu}(0,\tau) = R^{\mathrm{E}}_{uu}(u_0\tau, 0) \Leftrightarrow R^{\mathrm{E}}_{uu}(r,\tau) = R^{\mathrm{E}}_{uu}(r - u_0\tau, 0), \tag{4.16}$$

which has trivial consequences. If $r = u_0\tau$ it follows that $R^{\mathrm{E}}_{uu}(u_0\tau,\tau) = R^{\mathrm{E}}_{uu}(0,0)$, which means that along $r = u_0\tau$ the correlation function is constant and maximal. To test the validity of Taylor's hypothesis Favre et al. (1961) conducted laboratory experiments and calculated space-time correlations of the velocity fluctuations along an array of hot wire anemometers. The results showed a decrease of the correlation function along the line $r = u_0\tau$, which is an indication for temporal dynamics within the turbulent motion neglected in Taylor's hypothesis. He and Zhang (2006) introduced a new model for space-time correlations in turbulence called the elliptical model. Taylor's hypothesis is a first order approximation, whereas they use a second order approximation. The resulting isocorrelation lines are then ellipses and their shape is found to be influenced by two velocities, the convection velocity u_0 and some velocity shown to be equivalent to the sweeping velocity u_{sw}. Random sweeping appears to play an important role in the temporal decorrelation of turbulence.

The scale-dependent correlation function has been applied to HD simulations to isolate the effects that lead to decorrelation (Orszag and Patterson, 1972; Sanada and Shanmugasundaram, 1992). Similar to the space-time correlation functions random sweeping is found as the main decorrelator in HD turbulence (Sanada and Shanmugasundaram, 1992).

Space-time analysis of observations is a much more difficult topic in MHD compared to HD. Most plasma data is obtained from spacecraft measurements in the solar wind. Spacecraft are only single point measurements with negligible orbital velocity with respect to the solar wind speed in the sun's resting frame. The solar wind streams over the spacecraft with high speeds and therefore Taylor's approximation seems reasonable and has been used widely in the past. However, this is only an approximation and the spacecraft observation are inevitably a mixture of spatial and temporal fluctuations (Matthaeus et al., 2016; Perri et al., 2017). Only very recently Matthaeus et al. (2016) proposed a method to analyze space-time correlations from spacecraft data. Their approach is to transform correlations of single and multi-spacecraft measurements from the spacecraft frame into the solar wind frame moving with the mean solar wind speed $\mathbf{u}_0 = -u_0\mathbf{e}_r$.

To test the validity of Taylor's hypothesis they compare the spatial correlations $R(r,0)$ to the spatial correlations $R(v_{\mathrm{sw}}\tau,\tau)$. They find a significant difference between the two correlations and state that the normalized error between the two correlations increases with scale under consideration. The authors are inconclusive on the possible reasons for the underlying physics in the temporal decorrelation, but rather conclude that Taylor's hypothesis might not be a good approximation for upcoming spacecraft missions.



(a) Decorrelation times $\tau_D^{\pm}(k)$ (black and red) estimated from the scale-dependent correlation function $\Gamma_{\mathbf{z}^{\pm}}(k,\tau)$, best fits and the characteristic times $\tau_{\mathrm{nl}}(k)$, $\tau_{\mathrm{A}}(k)$, $\tau_{\mathrm{sw}}(k)$ and $\tau_d(k)$ as introduced in Section 2.2.4.

(b) Decorrelation time $\tau_D(k)$ (blue) estimated from the scale-dependent correlation function $\Gamma_{\mathbf{b}}(k,\tau)$ and $\tau_{\mathrm{sw}}(k)$ and $\tau_{\mathrm{nl}}(k)$ in the isotropic case $\mathbf{B}_0 = 0$.

Figure 4.1: Decorrelation times after (a) Servidio et al. (2011) and (b) Lugones et al. (2016).

Scale-dependent correlation functions have been applied to MHD simulations only very recently (Servidio et al., 2011; Lugones et al., 2016; Perri et al., 2017). Servidio et al. (2011) used isotropic ($\mathbf{B}_0 = \mathbf{0}$) MHD simulation and measure the decorrelation time $\tau_D(\mathbf{k})$, defined as the temporal lag τ where the scale-dependent correlation function decreases to $1/e$ of its initial value 1, $\Gamma(k,\tau_D) = 1/e$. The results are presented in Figure 4.1a together with the theoretical values of the non-linear time $\tau_{\mathrm{nl}}(k)$, the sweeping time $\tau_{\mathrm{sw}}(k)$, the Alfvén time $\tau_{\mathrm{A}}(k)$ and the dissipation time $\tau_d(k)$, estimated following the equations (2.44) - (2.47). The calculated correlation times of the Elsässer variables, $\tau_D^{\pm}$, are fitted and both scale with approximately $k^{-1.1\pm0.07}$ which is in favour of the random sweeping and Alfvén time, which both scale with k^{-1}. However, the authors do not state if the times shown in Figure 4.1a are only shifted for visualisation and if this discrepancy between the times and the measured decorrelation times is relevant. Lugones et al. (2016) also applied the scale-dependent correlation function to MHD simulations. They additionally analyzed anisotropic cases with increasingly large $\mathbf{B}_0$ to be able to distinguish between random sweeping and Alfvénic effects. Their results for the isotropic case support the study by Servidio et al. (2011) and they conclude for the non-linear time to be the time of decorrelation only for the largest wave numbers (see Figure 4.1b). As their definitions of the characteristic times are anisotropic with respect to the mean magnetic field, the Alfvén time is not shown for the isotropic case. They also do not consider the viscous diffusion time. According to Lugones et al. (2016) the random sweeping is more relevant than the Alfvénic interaction for moderate $\mathbf{B}_0$ and vice versa for larger $\mathbf{B}_0$, where τ_{A} decreases. They also state that the turbulent system selects the shortest correlation time available. Perri et al. (2017) use the scale-dependent correlation function to identify wave numbers, where random sweeping is not the relevant decorrelator, and conclude for this regime an invalidity of Taylor's hypothesis.

4.2.3 Lagrangian frame

The Lagrangian viewpoint is fundamentally different from the Eulerian viewpoint. No inertial frame is considered, but the properties along the trajectories of particles in the flow field are analyzed. A Lagrangian trajectory $\mathbf{x}(\mathbf{x}_0, t)$, evolving along a field $\mathbf{f}(\mathbf{x}(\mathbf{x}_0, t), t)$ is described by

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}, \tag{4.17}$$

where $\frac{d\mathbf{x}}{dt} = \frac{\partial}{\partial t}\mathbf{x} + (\mathbf{f} \cdot \nabla)\mathbf{x}$ is the Lagrangian derivative or total derivative and $\mathbf{f}(\mathbf{x}(\mathbf{x}_0, t), t)$ can be any field governing the flow. For MHD these are the velocity field $\mathbf{u}$ or the Elsässer fields $\mathbf{z}^{\pm}$, which both can be interpreted as momentum fluxes. The total derivative consist of the Eulerian derivative $\frac{\partial \mathbf{x}}{\partial t}$ at $\mathbf{x}$ and an advection or convection term $(\mathbf{f} \cdot \nabla)\mathbf{x}$.

The discretized scheme to obtain Lagrangian tracer trajectories $\mathbf{x}_k = \mathbf{x}(\mathbf{x}_0, t_k)$ from the non-kinetic Eulerian numerical simulation is given by

$$\mathbf{x}_k = \mathbf{x}_{k-1} + \Delta t \mathbf{f}(\mathbf{x}_{k-1}, t_{k-1}), \tag{4.18}$$

where Δt is the time increment so that $t_k = k\Delta t$. The Lagrangian field variables are technically replaced by the Eulerian fields at the specific time and space point in the simulation domain (Yu et al., 2012). That is often called a semi-Lagrangian scheme, as there are no true particles that are followed. For a true Lagrangian analysis a kinetic simulation or smoothed-particle simulation should be applied, but comes at a much larger computational cost (Price, 2012).

The Lagrangian two-point autocovariance function is defined as

$$C^{\mathrm{L}}_{\mathbf{vv}}(\mathbf{x}, t, \tau) = \langle \mathbf{v}(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x} + \mathbf{r}(t + \tau), t + \tau) \rangle, \tag{4.19}$$

where, in contrast to the Eulerian covariance function, the spatial lag $\mathbf{r}(t + \tau)$ depends on the temporal lag τ. The Lagrangian correlation function is thus only depending on the temporal lag τ. The lag $\mathbf{r}(t + \tau)$ can be considered the Lagrangian trajectory of the particle at starting point $\mathbf{x}$. A crosscovariance function may be defined analogously, but is not analyzed here. The relation to the Lagrangian energy density spectrum is given by

$$C^{\mathrm{L}}_{\mathbf{vv}}(\tau) = \int E^{\mathrm{L}}_{\mathbf{v}}(\omega) e^{i\omega\tau} d\omega. \tag{4.20}$$

Similar to the Eulerian frame the autocorrelation function is defined as

$$R^{\mathrm{L}}_{vv}(\tau) = \frac{\langle \mathbf{v}(\mathbf{x}, t) \cdot \mathbf{v}(\mathbf{x} + \mathbf{r}(t + \tau), t + \tau) \rangle}{\langle \mathbf{v}^2 \rangle}. \tag{4.21}$$

Application in HD and MHD turbulence

The analysis of Lagrangian correlations in hydrodynamics is not new (Durst et al., 1959; Deardorff and Peskin, 1970; Yeung and Pope, 1988; Kaneda, 1993; Shen and Yeung, 1997).

The tracing of particles in experiments has allowed this even before the availability of DNS. A main result of the Lagrangian correlation function in hydrodynamics is the difference between the decorrelation times in the Lagrangian frame and the Eulerian frame, the latter having a larger decorrelation time (Shen and Yeung, 1997).

In MHD no experimental particle tracing has been performed due to the complications in following a particle in a hot, confined laboratory plasma. The analysis of Lagrangian properties in MHD flows is thus limited to DNS and has only been debated in the literature very recently (Busse et al., 2007, 2010). However, no conclusions in addition to those of HD have been drawn from the Lagrangian correlation functions. However, the analysis of Lagrangian correlations often goes hand in hand with the analysis of dispersion of particle pairs. Some approaches to analyze behaviours of the dispersion or fluctuations of vector fields along the trajectories as well as previous results are introduced in the following.

4.3 Flow dispersion

The dispersion of particles in the flow field is a very different approach to statistically analyze the turbulent flow field. Although MHD turbulence theories predict a so-called elongated eddy, this schematic image cannot be transferred to the actual physical space. The distribution of energy in **k**-space leads to $k_\perp \gg k_\parallel$ for some MHD turbulence theories (Goldreich and Sridhar, 1995; Cho and Vishniac, 2000; Boldyrev, 2005). If naively transferred to physical space this means $l_\perp \ll l_\parallel$, the eddy scale being much larger parallel than perpendicular. However, this is not directly transferable. The relation of the actual physical structure of the flow field to the Eulerian statistics, where the distribution of energy in **k**-space originates from, is unclear (Kamps et al., 2009; Lalescu and Wilczek, 2017).

In this section we want to present novel techniques to analyze the anisotropy of the dispersion of test particles and how they help to identify driving effects in the flow dynamics. The analysis of distances between two particles with time is not new in HD research, in fact even older than Kolmogorov's work (Richardson, 1926; Durst et al., 1959; Deardorff and Peskin, 1970; Shen and Yeung, 1997; Yeung and Borgas, 2004). Some of the theory and results are presented in the first section of the chapter. Although anisotropy in the two particle dispersion of HD shear flows was observed (Shen and Yeung, 1997), it is poorly understood in the isotropic case. For the investigation of anisotropy in the dispersion of multi particle clouds the analysis of convex hulls is a novel approach introduced very recently by Pratt et al. (2017). In addition to the convex hull we introduce the principal component analysis (PCA) as a tool to measure anisotropy. Similar to the convex hull analysis (CHA) the PCA is not determined by a-priori knowledge of a reference direction along which anisotropy is measured. A-priori information is used when looking at the particle cloud as a pseudo-spheroid. Both the PCA and the pseudo-spheroid method have, to our knowledge, so far not been applied to flow dispersion.

4.3.1 Particle displacement

The displacement of two particles $\mathbf{x}_i$ and $\mathbf{x}_j$ is simply given by

$$\mathbf{r}(t) = \mathbf{x}_i(t) - \mathbf{x}_j(t), \tag{4.22}$$

where the initial separation is often denoted as $\mathbf{r}_0 = \mathbf{r}(t_0)$.

Based on the assumptions that the relative motions of particles are approximately constant for short times Batchelor (1950) derived a scaling of

$$\langle(\mathbf{r} - \mathbf{r}_0)^2\rangle \sim t^2, \tag{4.23}$$

where $\langle(\mathbf{r} - \mathbf{r}_0)^2\rangle$ is called the mean-squared relative displacement. Phenomenologically, this scaling can be associated to an advective process. The dispersion of two particles for short times in HD and MHD has been shown to scale with t^2 by several studies (see e.g. Shen and Yeung 1997; Busse et al. 2007).

Richardson (1926) assumed that the relative dispersion is directly proportional to the energy density spectrum. For HD he derived a scaling of

$$\langle\mathbf{r}^2\rangle \sim t^3, \tag{4.24}$$

which is often referred to as Richardson diffusion or Richardson law. A similar scaling was also found in the diffusion of magnetic field lines in MHD (Eyink et al., 2013). Servidio et al. (2016) find diffusion following Richardson's law in 2D kinetic simulations.

When the movement of the particles can be described by a diffusive process the mean-squared relative displacement is expected (Batchelor, 1950) to scale with

$$\langle(\mathbf{r} - \mathbf{r}_0)^2\rangle \sim t. \tag{4.25}$$

4.3.2 Convex hull analysis (CHA)

The convex hull is the smallest convex polygon that includes all points under consideration, e.g. a cloud of particles. An example of the convex hull (orange) for different shapes of particle clouds (blue points) is illustrated in Figure 4.2. The convex hull of a particle cloud can be used as an indicator of anisotropic dispersion (Pratt et al., 2017). For anisotropic dispersion the convex hull diverges from an approximately spherical shape that is expected if the particles disperse isotropically in space. If anisotropic dispersion is present, we expect the convex hull to no longer be of spherical shape (see Figure 4.2). Then, the ratio of the area S of the convex hull to its volume V increases, given a constant mean radius. To neglect the effect of the mean radius, where the volume increases with r^3 and the area with r^2, the ratio $S/V^{2/3}$ is a suitable parameter to study the anisotropy

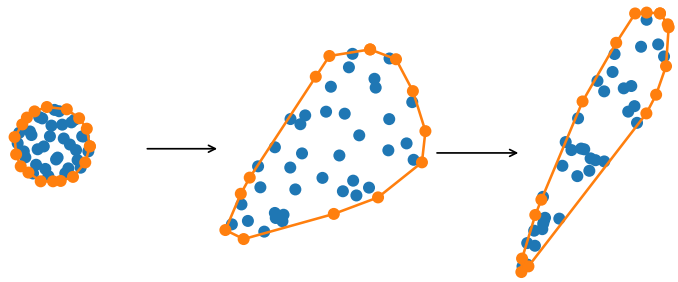


Figure 4.2: Illustration of a 2D cloud of particles dispersing in an MHD simulation with time from left to right. The connected particles (orange) resemble the convex hull.

of the cloud using the shape of the convex hull. If the convex hull is a perfect sphere this ratio is $(36\pi)^{1/3} \approx 4.84$. For simplicity the ratio is normalized by this factor.

The new tool was introduced only very recently by Pratt et al. (2017). They focused on the intermittency of particle dispersion within HD and MHD simulation and showed a significant spatial variation of the anisotropic dispersion within turbulence simulations.

4.3.3 Principal component analysis (PCA)

Principal component analysis (PCA) is a standard tool of multivariate statistics and has a wide application in data analysis (Pearson, 1901). It is often used for data clustering and image processing (e.g. Ding and He 2004). In a geometrical sense PCA determines the directions of highest variance within a point cloud, here a particle cloud in the flow field. There are several methods to compute the PCA. We introduce the most common method using the singular value decomposition (SVD) of the covariance matrix. First of all the cloud is demeaned by computing the distance vectors $\mathbf{d}_i(t_k) = \mathbf{x}_i(t_k) - \mathbf{x}_{\mathrm{com}}(t_k)$, where

$$\mathbf{x}_{\mathrm{com}}(t_k) = \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i(t_k), \tag{4.26}$$

can be regarded as the common center of mass of particles at position $\mathbf{x}_i(t_k)$ at time t_k. Now the covariance matrix is simply the outer product of the distance vectors

$$\mathbf{C}(t_k) = \sum_i \mathbf{d}_i(t_k)\mathbf{d}_i^T(t_k). \tag{4.27}$$

The SVD of $\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}$ then reveals the directions of highest variances, given by the singular vectors and values of $\mathbf{C}$. These directions are given by the columns of $\mathbf{U}$, whereas the non-zero singular values $\sigma_j, j = 1, 2, 3$ are the elements of the diagonal matrix $\mathbf{\Sigma}$. In a geometrical view the matrices $\mathbf{U}$ and $\mathbf{V}$ represent rotational matrices and $\mathbf{\Sigma}$ a scaling matrix. The direction of highest variance in this context can be interpreted as the direction of highest elongation of the particle cloud. To quantify the elongation of a point cloud

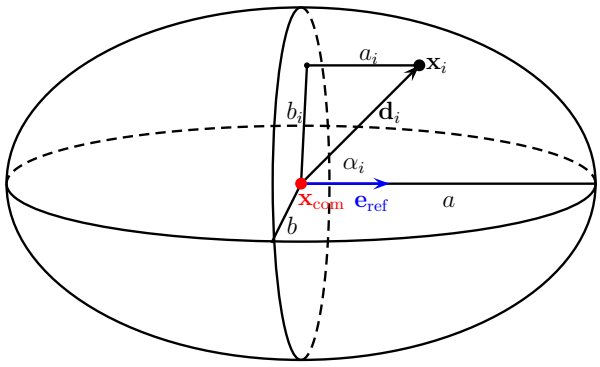


Figure 4.3: Schematic view of the pseudo-semi axes of a particle at $\mathbf{x}_i$ in a cloud represented as a spheroid with respect to the reference direction $\mathbf{e}_{\text{ref}}$.

we can compare the ratios of the highest variance, i.e. singular value σ_1, to the second highest, σ_2, as a function of time.

4.3.4 Pseudo-spheroid with a-priori semi axis

In order to analyze the anisotropic development of a cloud of particles it can be described as a spheroid

$$\frac{x^2 + y^2}{a^2} + \frac{z^2}{b^2} = 1, \tag{4.28}$$

where a and b are semi axes of the spheroid. The aspect ratio of the semi axes can be used to characterize the elongation of the cloud/spheroid along one of the axes. Within a cloud of particles one has to approximate these semi axes with respect to a reference direction $\mathbf{e}_{\text{ref}}$, e.g. a mean magnetic field. The distances, parallel and perpendicular to the reference direction $\mathbf{e}_{\text{ref}}$ can be calculated for the i-th particle $\mathbf{x}_i(t_k)$ at time t_k by

$$a_i = |\mathbf{d}_i| \sin(\alpha_i), \qquad b_i = |\mathbf{d}_i| \cos(\alpha_i), \tag{4.29}$$

where $\mathbf{d}_i = \mathbf{x}_i - \mathbf{x}_{\text{com}}$ and $\alpha_i = \cos^{-1}(\mathbf{d}_i \cdot \mathbf{e}_{\text{ref}}/(|\mathbf{d}_i||\mathbf{e}_{\text{ref}}|))$ (see Figure 4.3). The pseudo-semi axes a and b of the cloud are then defined as the mean of the distances a_i and b_i, respectively. The aspect ratio a/b (or similarly $1/N \sum_i a_i/b_i$) can be used to analyze the elongation of the cloud along the reference direction.

Here we use the mean field

$$\mathbf{f}_0(t_k) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{f}_i(t_k, \mathbf{x}_i(t_k),) \tag{4.30}$$

where $\mathbf{f} \in \{\mathbf{v}, \mathbf{b}, \mathbf{z}^{\pm}\}$, as our reference direction. As the particles disperse or evolve with time, the aspect ratio a/b is also a function of time.

5 Correlations in MHD simulations

In this chapter we want to apply the analysis methods from Section 4.2 to the incompressible MHD simulation introduced in Chapter 3. First, we use the classical spatial correlation function to quantify coupling between the flow fields and the forcing. To analyze temporal dynamics in the flow field we use space-time correlation in both the Eulerian and the Lagrangian framework. Also the scale-dependent correlation function is used to identify important decorrelation effects in the Eulerian framework.

5.1 Eulerian crosscorrelations between forcing and flow fields

Even though spatial Eulerian correlations have been studied extensively in the past the crosscorrelations between flow fields in MHD simulations are poorly understood or documented in the literature. We look at autocorrelations of the applied forcing, the velocity and the magnetic field, as well as the crosscorrelations between them.

Since the simulation domain is a $2\pi \times 2\pi \times 2\pi$ periodic domain we apply the correlation function with periodic boundary conditions. In Figure 5.1 the autocorrelation functions of the force field, the velocity and the magnetic field are shown. Naturally, the autocorrelation is 1 at $r_x = r_y = 0$ in all cases, which is in general not true for crosscorrelations. As the forcing is composed of π periodic sine and cosine functions with $k_f = 2$ the autocorrelation function shows the same periodicity. The correlation oscillates for a change of one spatial lag by $\pi/2$ between full correlation ($R^E_{FF} = 1$) and full anticorrelation ($R^E_{FF} = -1$) leading to a pattern in the correlation with the same periodicity as the forcing field (compare Figures 5.1a and 3.2). This is expected, as it can be derived analytically to be

$$R^E_{FF}(r_x, r_y, z) = \cos(k_f r_x)\cos(k_f r_y), \tag{5.1}$$

when considering only the xy-planes as a function of z (see Appendix A.1). The autocorrelations of the velocity and the magnetic field cannot be derived analytically. Here, we expect chaotic motion to decorrelate the fields. The autocorrelation functions of the velocity and the magnetic field are shown in Figure 5.1b and 5.1c, respectively. There is a significant difference between the two autocorrelations visible at first sight. Unlike the magnetic field the velocity shows a strong similarity to the autocorrelation function of

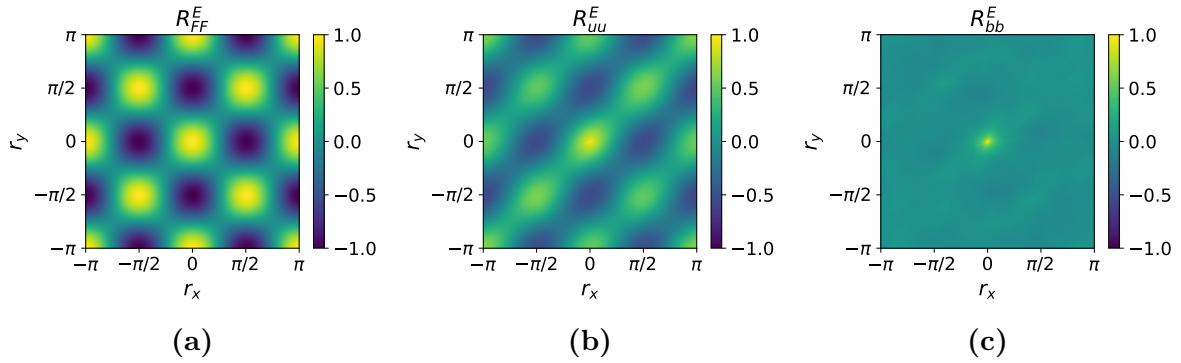


Figure 5.1: Autocorrelation functions R^E_{FF} (a), R^E_{uu} (b) and R^E_{bb} (c) in the xy-plane at $z = \pi$ of the force field $\mathbf{F^u}$, the velocity $\mathbf{u}$ and the magnetic field $\mathbf{b}$, respectively.

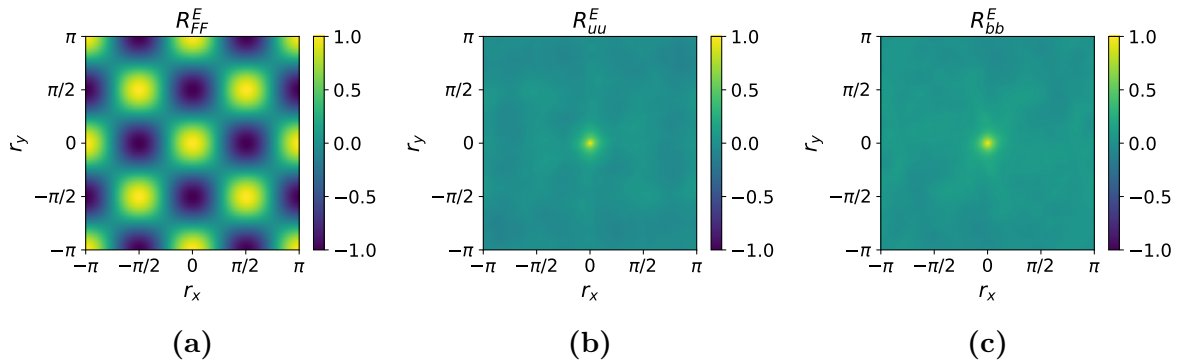


Figure 5.2: Autocorrelation functions R^E_{FF} (a), R^E_{uu} (b) and R^E_{bb} (c) in the xy-plane at $z = \pi/4$ of the force field $\mathbf{F^u}$, the velocity $\mathbf{u}$ and the magnetic field $\mathbf{b}$, respectively.

the forcing (compare Figures 5.1a and 5.1b). We expect this from the forcing acting only on the momentum equation (3.1). Less intuitive is the absence of the forcing pattern in the magnetic field autocorrelation shown in Figure 5.1c. The pattern can only be seen very slightly, with very low correlations ($R < 0.1$). Although the forcing solely acts on the momentum equation and is not induced specifically into the magnetic field one might expect the magnetic field to be coupled to the flow field under circumstances, where the frozen-in field theorem is applicable (see Section 2.2.1).

As the forcing is periodically changing from no forcing to maximal forcing as a function of z, the autocorrelation functions in a plane where the forcing vanishes, namely $z = \pi/4$, are also analyzed and presented in Figure 5.2. The autocorrelation of the forcing is, as expected from the analytical solution in equation (5.1), identical to the autocorrelation function at $z = \pi$ (compare Figures 5.1a and 5.2a). Unlike in the case of maximal forcing, the autocorrelation of the velocity does not show the periodic pattern when there is no forcing (compare Figures 5.1b and 5.2b). The autocorrelation of the magnetic field is visually indistinguishable between forcing and no forcing (compare Figures 5.1c and 5.2c). In both cases and for the velocity when no forcing is applied (see Figure 5.2a), the decorrelation is radially symmetric around $r_x = r_y = 0$ and underlines the global isotropy in the simulation.

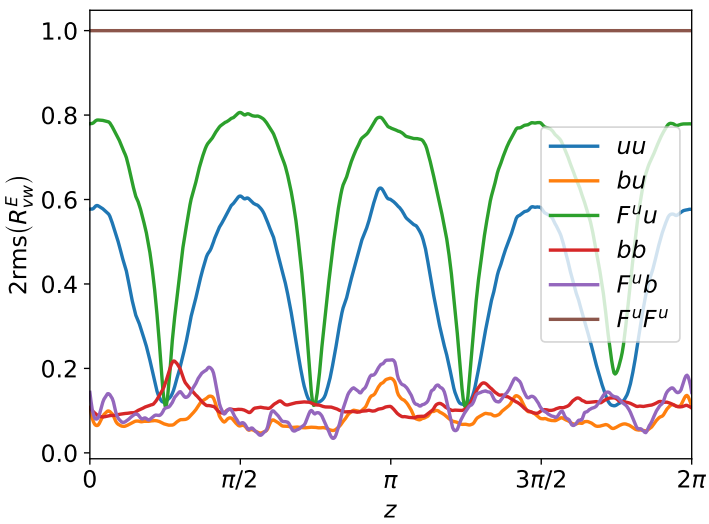
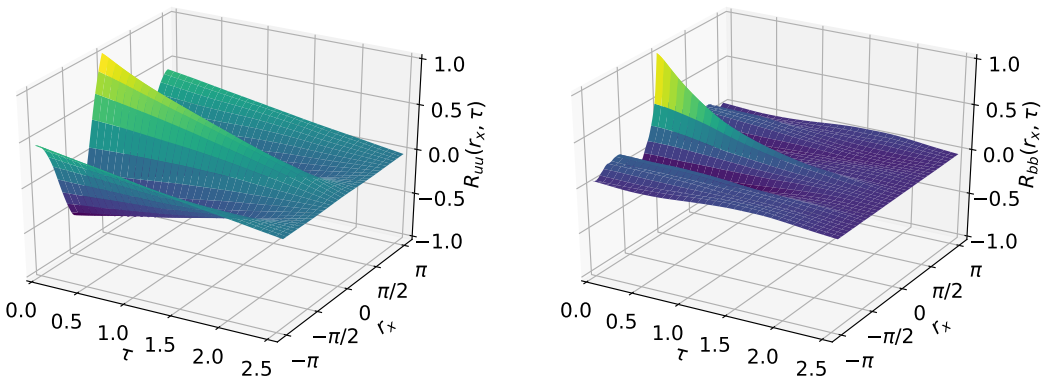


Figure 5.3: Rms of autocorrelations and crosscorrelations R^E_{vw} in the xy-plane along z between force field $\mathbf{F^u}$, magnetic field $\mathbf{b}$ and velocity $\mathbf{u}$.

As the autocorrelations of the fields indicate couplings between the forcing and the velocity, that are not visible between the forcing and the magnetic field, we want to analyze crosscorrelations between forcing, velocity and magnetic field next. Crosscorrelations allow a quantitative measure of the coupling of two fields. Further, to analyze the periodic coupling induced by the forcing not visually, but quantitatively, we simply take the rms values of the crosscorrelation functions, $\mathrm{rms}(R^E_{vw}(r_x, r_y, z))$. For the autocorrelation of the forcing we can derive analytically that $\mathrm{rms}(R^E_{FF}(r_x, r_y, z)) = 1/2$ for all z (see Appendix A.1). Therefore, we present in Figure 5.3 two times the rms of the crosscorrelations in the xy-plane as a function of z, so that $2\mathrm{rms}(R^E_{FF}(r_x, r_y, z)) = 1$ indicates the maximum coupling. The observations of the rms as a function of z underline the indications of the autocorrelation functions. The crosscorrelation between the forcing and the velocity changes periodically with the period of the forcing amplitude (see Figure 5.3, green line). It reaches a maximum of 0.8, when the forcing is maximal and decreases to ≈ 0.15 when the forcing is off. The dominance of the forcing in the flow field is apparent. Also the presence of non-linear interactions in the velocity flow field, leading to decorrelation, can be concluded. The magnetic field, on the other hand, is completely uncorrelated to the forcing. No periodicity is observed in the rms of the crosscorrelation between magnetic field and forcing (see Figure 5.3, violet line). Likewise, no correlation between magnetic field and velocity is observed, regardless of additional forcing present or not (see Figure 5.3, orange line). This is not only valid for the rms of the crosscorrelation between magnetic field and velocity, but also for the crosscorrelation without any spatial lag, which are not shown here. They show the same behaviour for all crosscorrelations under consideration as the rms shown in Figure 5.3. The missing correlation between the velocity and magnetic field without any spatial lags indicates that they are decoupled and evolve uncorrelated. The magnetic field does not seem to be frozen into the flow. The simulation is non-ideal



(a) **(b)**

Figure 5.4: Space-time autocorrelations of (a) the velocity and (b) the magnetic field.

MHD, which, together with the large amount of turbulent motion, renders the frozen-in assumption invalid (Aluie, 2009; Eyink et al., 2013).

The spatial crosscorrelations give insight into the coupling between forcing, velocity and magnetic field, but since the analysis is purely spatial, no information on the effect leading to energy transfer and the decay of eddies is given. We therefore want to investigate space-time correlations. In Figure 5.4 the space-time autocorrelations of the velocity and the magnetic field are shown along a sample line at $y = z = \pi$ and $x = [0, 2\pi]$ for the whole simulation time. First of all, the periodicity in the spatial domain induced by the forcing is again recognizable in the velocity correlation. The decorrelation in space is again much slower for the velocity than the magnetic field. The temporal decorrelation of the velocity and the magnetic field both decay to 0 at the maximum time available, $T = 2.56$, due to the finite time series. The difference between velocity and the magnetic field in the temporal decorrelation is observed in the early time lags. The correlation of the velocity decays almost linearly, indicating that there is negligible temporal decorrelation and the decay of the correlation function is solely due to the zero padding of the finite time series. The magnetic field on the other hand decays faster and not at all linearly. There, a non-trivial effect is expected to decorrelate. As the space-time correlations are dominated by the large scale motions and, for the velocity, this is mainly the constant forcing so that no clear conclusions can be drawn on the decorrelation effects present in the flow. Therefore, we want to analyze the correlation scale-dependently to isolate decorrelations on a specific fluctuation/eddy scale "smeared out" by the large scale fluctuations.

5.2 Scale-dependent Eulerian correlation function

To distinguish between large scale and local effects the space-time correlation function is insufficient. The scale-dependent correlation function introduced in Section 4.2.2 allows the analysis at different wave numbers corresponding to the fluctuations at a certain scale.

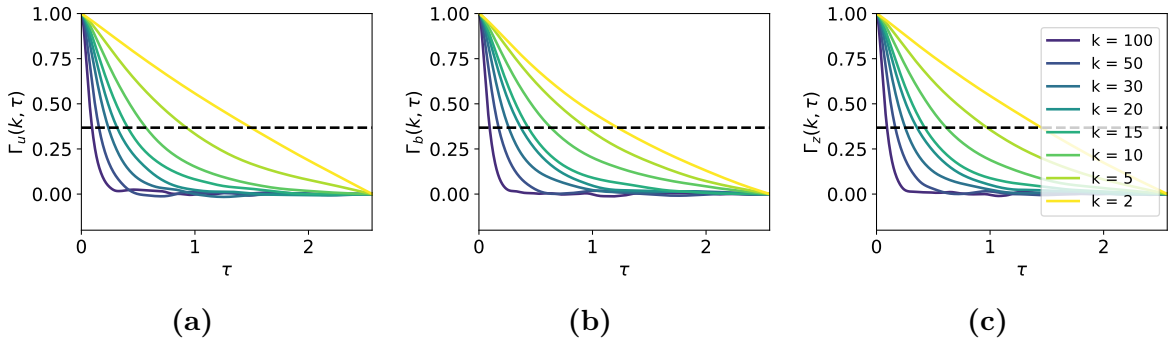


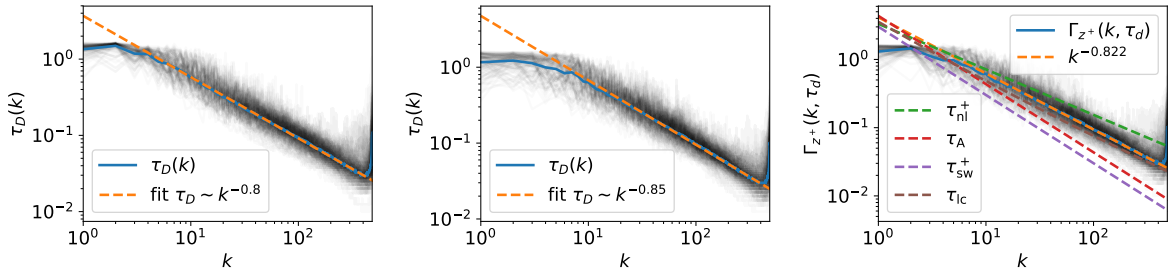
Figure 5.5: Averaged scale-dependent decorrelation functions $\Gamma_{\mathbf{w}}(k,t)$ of the velocity (a), the magnetic field (b) and the positive Elsässer variable (c). The samples are taken for multiple slices through the simulation domain and the average is taken between these samples. Wave numbers are chosen arbitrarily and listed in (c).

We apply the correlation function, given in equation (4.13), to multiple cuts through the simulation domain, in areas with high forcing as well as no forcing. The average scale-dependent correlation function $\Gamma_{\mathbf{w}}(k,\tau)$ is shown in Figure 5.5 for $\mathbf{w} = \mathbf{u}, \mathbf{b}, \mathbf{z}^+$. Wave numbers have been selected arbitrarily between the wave number on which forcing is induced, $k_f = 2$, and wave numbers smaller than 100, approximately representing the upper end of the inertial range (see Figure 3.1).

All three scale-dependent correlation functions displayed in Figure 5.5 show similar properties. Since finite time series are used they all decay towards zero for $t \to 2.56$. The scale-dependence is similar for all fields under consideration. Correlations of higher wave numbers, corresponding to smaller scales, decay faster in time than smaller wave numbers associated with larger scales. When explaining the scale-dependence with the decay of eddies of a certain scale we expect a shorter decorrelation time for smaller scales whose decay time is smaller, which is consistent with the observations. The most prominent difference between the three fields is the decorrelation at the smallest wave number $k = 2$. Similar to the scale-independent space-time correlation, the decay is almost linear for the velocity (compare Figures 5.5a and 5.4a). The reason for this is the time constant forcing applied to the flow field, which is not true for the magnetic field decorrelation at $k = 2$ (see Figure 5.5b). For the Elsässer variable z^+ the decorrelation at $k = 2$ is in between the linear decorrelation of the velocity and the decorrelation of the magnetic field. The scale-dependent correlation function clearly allows the isolation of local fluctuations in the flow field and separate large scale effects, e.g. the forcing, that dominate the decorrelation at smaller scales.

To distinguish between relevant effects in the decorrelation we want to compare measured decorrelation times with characteristic times in the flow, namely those introduced in Section 2.2.4. We assume an exponential decay of the correlation function, so that

$$\Gamma(k,\tau) \sim e^{-\tau/\tau_D(k)}, \tag{5.2}$$



(a) **(b)** **(c)**

Figure 5.6: Scale-dependent decorrelation times of the velocity (a), the magnetic field (b) and the Elsässer variable (c). The black lines show each sample. The blue lines show the averaged decorrelation times. The dashed orange lines show a least squares fit to the averaged decorrelation times for wave numbers $10 < k < 400$. In (c) the characteristic times after Servidio et al. (2011) are shown, as well as their arithmetic mean τ_{lc} (brown).

where $\tau_D(k)$ is the scale-dependent decorrelation time. Fitting the correlation function to this model is impractical as large errors may occur due to the zero padding from the finite time series. Here, also due to comparability with earlier studies by Servidio et al. (2011) and Lugones et al. (2016), we simply define the decorrelation time to be

$$\tau_D(k) = \tau\big|_{\Gamma_w(k,\tau)=1/e}. \tag{5.3}$$

The horizontal dashed lines in Figure 5.5 indicate $1/e$. The measured decorrelation times $\tau_D(k)$ are presented in black in Figures 5.6a, 5.6b and 5.6c for the velocity $\mathbf{u}$, the magnetic field $\mathbf{b}$ and the Elsässer variable $\mathbf{z}^+$, respectively. The average of all samples is shown as a blue line. The least squares fit to the correlation times for $10 < k < 400$ are hown in orange dashed lines. When approaching $k = 482$ the decorrelation times increase sharply, which is due to aliasing in the simulation. Another clear feature is the peak at $k = 2$ for the velocity and the Elsässer variable caused by the forcing, which is not found in the magnetic field correlation functions. In all three cases the variation in the samples is large, whereas the average is remarkably steady. The average shows a power-law for wave numbers between 10 and 400 with the least squares fit to τ_D matching the decay almost perfectly. The power-law shows values of $-0.8 \approx -4/5$ for the velocity field, -0.85 and -0.82 for the Elsässer field, where the fit errors are negligibly small. The system does not seem to decorrelate with the fastest decorrelation time available, the Alfvén time $\tau_\mathrm{A} \sim k^{-1}$ or the random sweeping time $\tau_\mathrm{sw} \sim k^{-1}$. The estimations of the characteristic times are shown in Figure 5.6c. It is seen that the shortest time is the random sweeping time (violet), followed by the Alfvén time (red) and the slowest is the non-linear time (green). If we assume that the decorrelation time scales as the linear combination of all characteristic times, a prediction, τ_{lc}, for the decorrelation time can be made from the theoretical times. The estimation τ_{lc} is shown in Figure 5.6c in brown,

having slightly shorter times than the measured decorrelation times. Also, the coefficients $C_{-1} = (C_{\mathrm{sw}} + C_{\mathrm{A}})$ and C_{nl} of the characteristic times in a linear combination

$$\tau_D(k) \sim \tau_{\mathrm{sw}}(k) + \tau_{\mathrm{A}}(k) + \tau_{\mathrm{nl}}(k) \sim (C_{\mathrm{sw}} + C_{\mathrm{A}})k^{-1} + C_{\mathrm{nl}}k^{-2/3}, \tag{5.4}$$

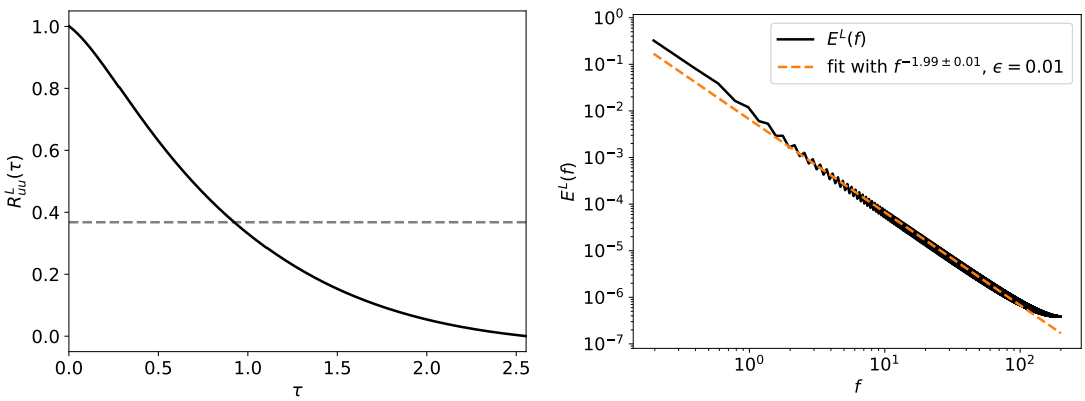
can be estimated by least-squares fitting. A further distinction between C_{sw} and C_{A} is not possible as both the sweeping and the Alfvén time scale with k^{-1} and an anisotropic analysis is needed (Lugones et al., 2016). For the velocity the coefficents are estimated to be $C_{-1} = 2.88 \pm 0.05$ and $C_{\mathrm{nl}} = 1.34 \pm 0.02$. For the magnetic field the coefficients are $C_{-1} = 3.56 \pm 0.09$ and $C_{\mathrm{nl}} = 1.29 \pm 0.03$ and for the Elsässer variable the coefficients are $C_{-1} = 3.1 \pm 0.07$ and $C_{\mathrm{nl}} = 1.36 \pm 0.02$. We observe an almost constant coefficient of the non-linear time between the different mean fields, whereas the C_{-1} coefficient increases from velocity to Elsässer field to magnetic field. An intuitive interpretation is the increasing importance of the Alfvén time τ_{A} when including the magnetic field fluctuations. However, an increase of random sweeping and Alfvénic interactions is ambiguous.

5.3 Lagrangian correlations in MHD simulations

The results from the Eulerian space-time correlation function and the scale-dependent correlation function are inconclusive on what is a dominant phenomenon in the decorrelation of an MHD flow field. The Lagrangian trajectory can follow the velocity $\mathbf{u}$ or the Elsässer field $\mathbf{z}^{\pm}$ which both resemble momentum transport.

We trace 5000 particles through the simulation along the mass flux $\mathbf{f} = \mathbf{u}$ and calculate the correlation function following Section 4.2.3. The Lagrangian autocorrelation function of the velocity is shown in Figure 5.7a. The decorrelation time τ_D is estimated to be approximately 0.93. This is shorter than the kinetic and magnetic large eddy turnover times and shorter than the Eulerian decorrelation time, which is estimated to be approximately 1.29. It indicates that the Lagrangian correlation function shows different properties of the flow, compared to the Eulerian correlation function, which is mainly dominated by the largest scales. The Lagrangian correlation function of the magnetic field and the Elsässer field are similar to the velocity correlation.

The Lagrangian covariance function is directly related to the Lagrangian energy density spectrum $E_w^{\mathrm{L}}(\omega)$ by equation (4.20) and is shown in Figure 5.7b. Similar to the Eulerian frequency spectrum (not shown) no turbulent cascade is visible, which is due to the fact that the simulation time is only just longer than one large eddy turnover time. The energy injection frequency is thus not visible. The shallowing of the spectrum towards frequencies $f > 100$ could be explained by the fact that only every 10th integration step is saved in this simulation and frequencies $f > 1/\delta t = 400$ could contribute to these frequencies by aliasing. The slope of the Lagrangian spectrum is -1.99 ± 0.01, which is in agreement with theoretical predictions and previous studies observing a -2 spectrum (Busse et al., 2010). All theories for Lagrangian spectra result in the same scaling of



(a) **(b)**

Figure 5.7: Average Lagrangian autocorrelation of the velocity (a) and the corresponding energy density spectrum (b).

$C_L f^{-2}$ and the different mechanisms or assumptions used in the derivation only result in a different C_L (Busse et al., 2010). Here, where the simulation time is not large enough to have a reliable estimate of the energy injection regime, an estimation of C_L is unwise and we are unable to distinguish between the theoretical predictions.

Since the Lagrangian correlation is purely time dependent, no scale-dependent Lagrangian correlation function exists, at least not to the author's knowledge. Thus, it is impossible to distinguish between decorrelation on different eddy scales from the Lagrangian correlation function. A possibility to analyze properties of particles along Lagrangian trajectories scale-dependently is introduced in the following Chapter.

5.4 Discussion

Correlation analysis was performed to quantify the influence of the forcing in an MHD simulation on the velocity field and the magnetic field. The Taylor-Green forcing acts solely on the momentum equation. The crosscorrelation function revealed that the forcing explicitly acts on the velocity field and no sign of influence on the magnetic field is found. The magnetic field showed no pattern associated with the forcing, in contrast to the velocity. The rms of these crosscorrelations underlines this conclusion as it characterizes the overall coupling between the fields on any spatial lag and no sign of coupling between magnetic field and forcing was found. For maximum forcing the coupling between velocity and forcing reaches up to 0.8, where 1 would be perfect coupling. The difference from the perfect coupling is accounted to the turbulent flow motion. The missing correlation between velocity and magnetic field indicates that the magnetic field cannot be considered frozen into the flow. The turbulence, giving rise to reconnection processes, breaks the assumptions necessary for the frozen field theorem to hold (Alfvén, 1943). This is in agreement with previous conclusions using the same simulation data (Eyink et al., 2013).

As the (space-time) correlations of the velocity are largely influenced by the forcing, a scale-dependent analysis was applied to filter out the large scale correlations. The scale-dependent Eulerian correlation function showed a faster decorrelation of small scale fluctuations than of large scale fluctuations. The forcing is also clearly recognizable in the decorrelation times of the velocity and the Elsässer field at the injection scale $k = 2$. Again, no sign of forcing in the magnetic field. As the characteristic times present in MHD, namely the non-linear time $\tau_{\mathrm{nl}} \sim k^{-2/3}$, the Alfvén time $\tau_{\mathrm{A}} \sim k^{-1}$ and the random sweeping time $\tau_{\mathrm{sw}} \sim k^{-1}$ are also functions of wave number k the decorrelation time τ_D obtained from the scale-dependent correlation functions is a good indicator for the dominant processes in the flow. The decorrelation time decreases almost perfectly in a power law for scales below the energy injection regime. For the velocity τ_D decays with $k^{-0.8}$, whereas for the magnetic field and the Elsässer field τ_D decays with $k^{-0.85}$ and $k^{-0.82}$, respectively. The system does not seem to select the shortest decorrelation time available. We proposed the decorrelation time decays proportional to the superposition of all characteristic times. The results of the least squares fit of such a superposition show an almost constant contribution of the non-linear time for all mean fields. An increased contribution of the characteristic times scaling with k^{-1}, namely the Alfvén time and the random sweeping time, to the decorrelation time from velocity to Elsässer variable to magnetic field is observed. Both an increased importance of the Alfvén time in the magnetic field fluctuations or larger sweeping induced by the large scale magnetic field, which cannot be removed by Galileo transformation, are possible explanations. The linear combination of theoretically estimated characteristic times agrees with the observed decorrelation times. The results presented are not strictly in agreement with the literature, where the common conception is a dominance of random sweeping in the decorrelation of turbulence without a large mean magnetic field (Servidio et al., 2011; Lugones et al., 2016). They base the conclusion on the argument that the random sweeping time is the shortest characteristic time in the system, in agreement with our estimation of τ_{sw}. An interesting feature of the scale-dependent correlation function is the absence of any changes in the slope of the decorrelation times for wave numbers larger than the upper end of the inertial range, $k = 50$. The decorrelation times do not seem to have a large coupling to the wave number spectrum. Servidio et al. (2011) also observed no "spectral break" in the decorrelation times, but do not give any explanation. Perri et al. (2017) observe a different decay of τ_D at scales below the inertial range, but give no explanation. They use this change in the decorrelation behaviour as an indication for the invalidity of Taylor's approximation.

Another way of analysis is the Lagrangian viewpoint. We traced pseudo-particles through the simulation domain and analyzed the Lagrangian correlation functions. The decorrelation time is found to be shorter than the Eulerian correlation time, which is in agreement with previous observations (Busse et al., 2007). The Lagrangian covariance function is directly related to the Lagrangian energy density spectrum (shown in Figure

5.7b). It scales with f^{-2}, which is in agreement with theoretically derived spectra (Busse et al., 2010), but it is unclear if the visible inertial range is compromized by the limited length of simulation time available. Due to the fact that the simulation time is only one kinetic large eddy turnover time the energy input regime is not visible. Also no sign of a spectral break is found. The power at high frequencies shows a shallower slope than f^{-2}, which is unexpected, but could be explained by aliasing occurring from the fact that only every 10th time integration step is saved in the simulation. As the Lagrangian energy density seems to be compromized by the lack of time and resolution, further conclusion on the spectrum are misguiding.

6 Anisotropic dispersion of semi-Lagrangian particles in MHD

To investigate possible phenomena in the dispersion of Lagrangian particles in MHD we trace clouds of particles on different scales and apply the methods introduced in Section 4.3. The shape (or the properties describing the shape) of the dispersing particle clouds could give insight into the underlying processes.

We investigate 100 clouds with 50 particles each initially on spheres of different radii, which determine the scales. The points are approximately equally distributed on the sphere following a simple algorithm based on the spherical Fibonacci projection to avoid any averaging biases (Swinbank and Purser, 2006). The initial cloud centers are chosen randomly in the $2\pi \times 2\pi \times 2\pi$ domain. All pseudo-particles of the clouds are then traced in the flow field following the mass flux so that $\mathbf{f} = \mathbf{u}$ in the Lagrangian propagation given by equation (4.17). Each cloud is statistically regarded as a trial and for the analysis of different quantities averages are calculated over all clouds/trials in the following.

6.1 Particle dispersion

First of all we analyze the dispersion of the particles simply by looking at the mean displacements between the particles. Since not only two particles are considered, the displacement is defined for each particle with respect to the center of mass of the cloud, so that

$$\mathbf{r}_i(t) = \mathbf{x}_i(t) - \mathbf{x}_0(t). \tag{6.1}$$

The initial displacement is given by $\mathbf{r}_{i,0} = \mathbf{r}_i(t_0)$. The results for this approach have to be normalized by the initial separation $\|\mathbf{r}_0\|$ corresponding to the scale of the cloud k^{-1}.

We compute the mean squared relative displacement normalized by the wave number, $\langle(\mathbf{r}(t) - \mathbf{r}_0)^2\rangle k$, for all scales as a function of time. The results are presented in Figure 6.1. For comparison with previous studies the time is normalized by the kinetic Kolmogorov time scale $\tau_{\eta,u} = (\nu/\varepsilon_u)^{1/2} \approx 0.1$. The mean squared relative displacements show a clear t^{-2} scaling expected from Batchelor (1950) and in agreement with Busse et al. (2007). The scale-dependence is compensated for by the wave number k. Without the normalisation factor the scale-dependence is found to be similar to the results of Busse et al.

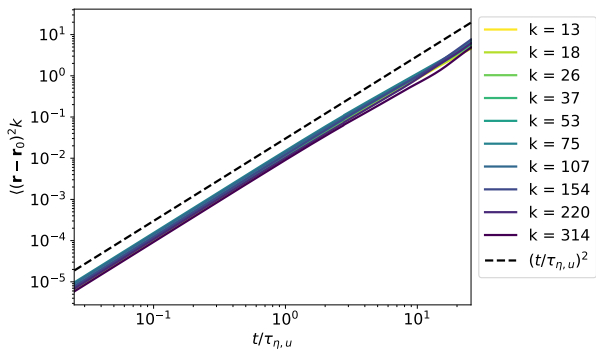


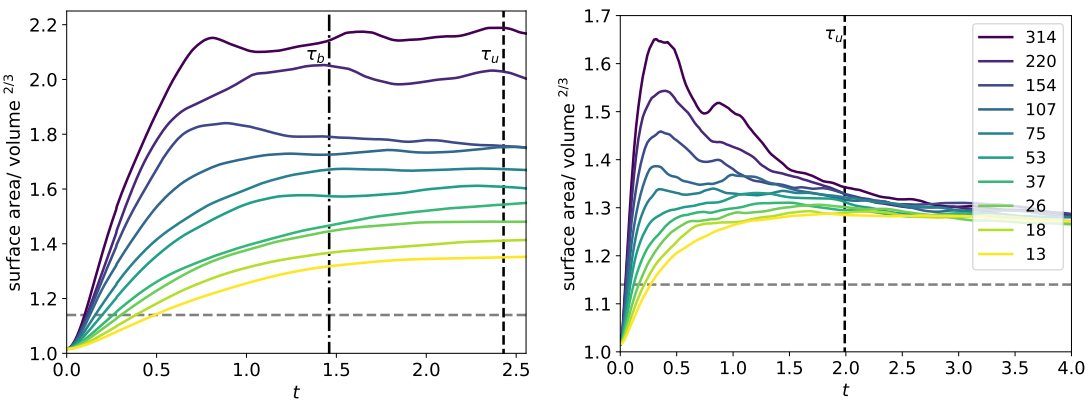
Figure 6.1: Diffusion of particles in MHD for the different cloud sizes. The time shown is normalized by the kinetic Kolmogorov time scale $\tau_{\eta,u} = 0.1$.

(2007), where larger initial separations show a higher mean squared relative displacement compared to smaller initial separations. In contrast to Busse et al. (2007) there is no change in the slope of $\langle(\mathbf{r} - \mathbf{r}_0)^2\rangle k$, probably due to the fact of the limited simulation time. Almost no changes are observed by Busse et al. (2007) for $t/\tau_{\eta,u} < 20$, which is the maximum time available here.

6.2 Convex hull analysis

The convex hull of the particle clouds is the first property we use to analyze the anisotropy of the dispersion. The initial cloud is a sphere and thus the ratio of the area to the volume$^{2/3}$ is equal to 1 when normalizing by the factor $(36\pi)^{1/3}$. The ratio cannot decrease below 1 and we expect it to increase when following the cloud in the simulation as the particles do not disperse on a perfect sphere. A threshold is determined, to infer beyond which the ratio shows significant anisotropy. We take the natural deviation from the perfect sphere when a cloud disperses isotropically, where each particle follows a Brownian motion, as our threshold. It depends on the number of particles on the initial sphere. For 50 particles, used here, the average ratio is estimated via Monte Carlo method to be ≈ 1.14 and constant in time (see Appendix A.2). We consider anything above this threshold to be significantly anisotropic.

At every time step t the convex hull of each cloud and the normalized ratio of the area to the volume is calculated. The ratios are averaged over all clouds at each time step. For comparison the analysis is repeated for HD. The results for MHD and HD are presented in Figure 6.2a and Figure 6.2b, respectively. The grey dashed horizontal line at a ratio of 1.14 indicates the threshold above which anisotropy is significant. The ratios are shown as averages over all trials at one scale. Each scale is indicated by color from small scales $k = 314$ (blue) to large scales $k = 13$ (yellow). For all scales an increase of the ratio from the initial condition is observed for early times, as expected. For late times the increase of the ratio falls of and in MHD even saturates for smaller scales (see Figure 6.2a, blue lines). The most dominant difference between the scales is the amplitude of



(a) **(b)**

Figure 6.2: Normalized average ratios of the area to the volume$^{2/3}$, $\langle S/V^{2/3}\rangle(36\pi)^{-1/3}$, in (a) MHD and (b) HD. The colors of the wave numbers k are listed in the legend in (b). The horizontal grey line indicates threshold value and the kinetic large eddy turnover times τ_u are shown as vertical dashed lines. For MHD the magnetic large eddy turnover time τ_b is also shown.

the ratios when evolving in time. It is observed that the increase of the ratio steepens with decreasing scale. The ratios of small scales reach up to higher ratios (e.g. ≈ 2.1 for $k = 314$) than those of large scales (e.g. ≈ 1.3 for $k = 13$). Thus, clouds of initially smaller scale, i.e. radius of the initial sphere, disperse more anisotropically than clouds of larger scales. For times $\tau \to \infty$ we would expect the ratio to converge towards the threshold value, as the increasingly separated particles would eventually flow fully independently and the relative movement is governed by a diffusive process. Unfortunately we only have a simulation time slightly longer than the kinetic large eddy turnover time $\tau_u = 2.43$ and any long term evolution remains uncertain.

To identify which effects in the evolution of the convex hull are unique to MHD we want to compare it with the same analysis applied to an HD simulation. The results are shown in Figure 6.2b. Again, an expected increase, which is also scale-dependent, of the ratio for early times similar to MHD occurs. Although, in terms of large eddy turnover time τ_u, the simulation time is twice as long for HD compared to MHD, the difference to MHD is clear. The ratios for HD increase scale-dependently up to a ratio of ≈ 1.65 at time $t \approx 0.3$ for the smallest scale, which is around 1/3 less than the ratio for the smallest scale in MHD. From their maximum at $t \approx 0.3$ they decrease in such a manner that the ratios of all scales approximately converge to the same value of ≈ 1.3 at roughly the large eddy turnover time τ_u. At times larger than τ_u all scales show almost the same ratio which slowly decreases with time. We expect this decrease to converge towards the threshold value. MHD flows seem to sustain scale-dependent anisotropy for a longer period of time than HD flows, where scale-dependence is lost at the large eddy turnover time.

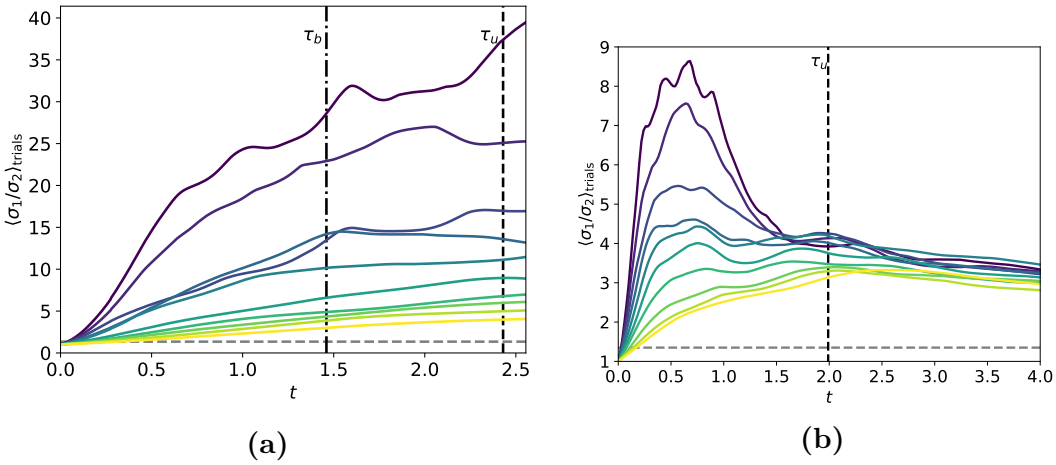


Figure 6.3: Average ratios of the highest to second highest singular value, $\langle\sigma_1/\sigma_2\rangle_{\text{trials}}$, of the clouds in (a) MHD and (b) HD.

6.3 Principal component analysis

The CHA revealed anisotropic dispersion for both HD and MHD flows, where the latter seems to sustain the anisotropy for a longer period. The CHA does not, however, reveal the shape of the cloud. Whether it takes a cigar shape, a discus shape or any other shape is unclear (Pratt et al., 2017). To validate the findings of anisotropic dispersion and to get further insights into the shape we apply the PCA to both MHD and HD.

We follow the PCA method described in Section 4.3.3 to quantify the largest variances in particle separations. The average ratios of the largest to the second largest variances (singular values) are calculated. The results are shown in Figures 6.3a and 6.3b for MHD and HD, respectively. The ratios in both cases show properties similar to the CHA. Except for very early times the ratios are well above the threshold value of 1.36 calculated for isotropically dispersing clouds (see Appendix A.2). For early times all ratios increase scale-dependently from the initial value of 1. Also similar to the convex hull for HD the scale-dependence is lost at roughly $t = \tau_u$ (compare Figures 6.3b and 6.2b). However, the time at which the ratio reaches a maximum for small scales, $t \approx 0.6$, is approximately twice as large compared to CHA. The slow decay for late times is again observed for HD. The PCA is also similar to the CHA for MHD. As in the CHA the ratio values are larger than the ratios for HD. The maximum of the smallest scale reaches a ratio of ≈ 16 compared to ≈ 8.5 for HD. The MHD ratios in PCA are also longer scale-dependent compared to the HD ratios, similar to CHA. It is, however, less clear if they saturate the same way as in the CHA. The smallest scales reach their maximum value at times larger than τ_b, whereas in the CHA this is the case for $\gtrapprox \tau_b/2$ (compare dark blue lines in Figures 6.3a and 6.2a). Besides the slight differences in maxima and saturations both CHA and PCA are in agreement and seem to capture the same properties of the anisotropic behaviour in the MHD and HD flows.

As an addition to the CHA, the PCA allows the differentiation of the shapes of the clouds into cigar and discus shapes. For a cigar shape the largest variance σ_1 is much larger than the second largest σ_2 and the smallest variance σ_3, which are approximately equal, $\sigma_2 \gtrapprox \sigma_3$. For a discus shape the first two variances are approximately equal, so that $\sigma_1 \gtrapprox \sigma_2$ and $\sigma_2 > \sigma_3$. We separate the clouds into the two groups of cigar shapes ($\sigma_1 > \sigma_2 \gtrapprox \sigma_3$) and discus shapes ($\sigma_1 \gtrapprox \sigma_2 > \sigma_3$) and find that both are abundant in MHD and HD. There is no preferred shape, all shapes contribute in a comparable manner to the average. In case the elongation of eddies was directly transferable to the cloud dispersion one would expect the particles to disperse in a cigar shape azimuthally symmetric along the mean magnetic field direction. Neither the direction of highest variances and the mean magnetic field direction coincide in general, nor does the mean flow velocity. There is also no observable difference in the anisotropy of the dispersion between clouds of different initial cross-helicity $\mathbf{B}_0(t_0) \cdot \mathbf{u}_0(t_0)$.

6.4 Pseudo-spheroid analysis

Both the CHA and the PCA do not use a-priori information. The anisotropy of the cloud is analyzed with no pre-defined reference direction. For MHD there are a few reference directions that are of interest and could have an influence on the dispersion. Besides the mean velocity direction $\mathbf{v}_0(t)$, also present in HD, the mean magnetic field $\mathbf{b}_0(t)$ and the mean Elsässer fields $\mathbf{z}^{\pm}(t)$ are characteristics in MHD flows. These mean directions can be computed at any time as an average over all points in one cloud. The distance vectors with respect to the mean position are projected on this reference direction and the parallel and perpendicular distances are then computed accordingly (see Section 4.3.4). Analogue to CHA and PCA a threshold value for the ratio of the pseudo-semi axes is estimated to be $a/b \approx 1.44$. The pseudo-semi axes ratio is computed with $\mathbf{v}_0(t)$ as the reference direction for MHD and HD, shown in Figures 6.4a and 6.4b, respectively. For MHD the ratio is also computed with respect to $\mathbf{b}_0(t)$ shown in Figure 6.5a and with respect to $\mathbf{z}_0^{+}(t)$ shown in Figure 6.5b (the negative Elsässer variable is similar to the positive). Different to the ratios of CHA and PCA the pseudo-semi axes ratio can decrease from its initial value of 1 for a perfect sphere. For some scales at very early times in MHD and even for very late times in HD the ratio is found to be below 1 (Figure 6.4b). For HD the pseudo-semi axes ratio is low at all scales and all times. No clear scale dependence is recognizable and the values are well below the significant threshold value of 1.44. Therefore, no anisotropic dispersion along the mean flow direction $\mathbf{v}_0(t)$ is detected with this method for HD.

The MHD pseudo-semi axes ratios show scale-dependence for all reference directions. Clouds of initially small scales have a higher ratio compared to those with initially large scales, similar to the scale-dependence found in the previous CHA and PCA. Except for the largest scales the ratios surpass the threshold value of 1.44 after a certain time. For the smallest scale this is approximately around 0.3–0.5 for all reference directions (see

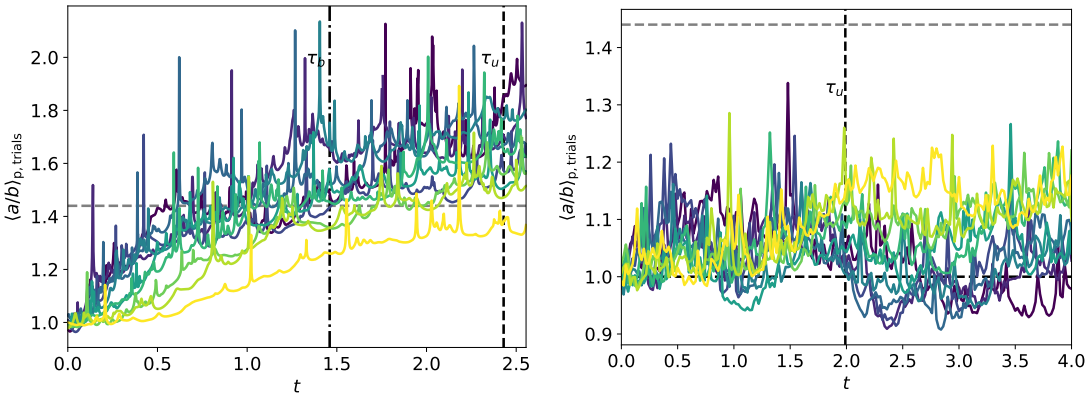


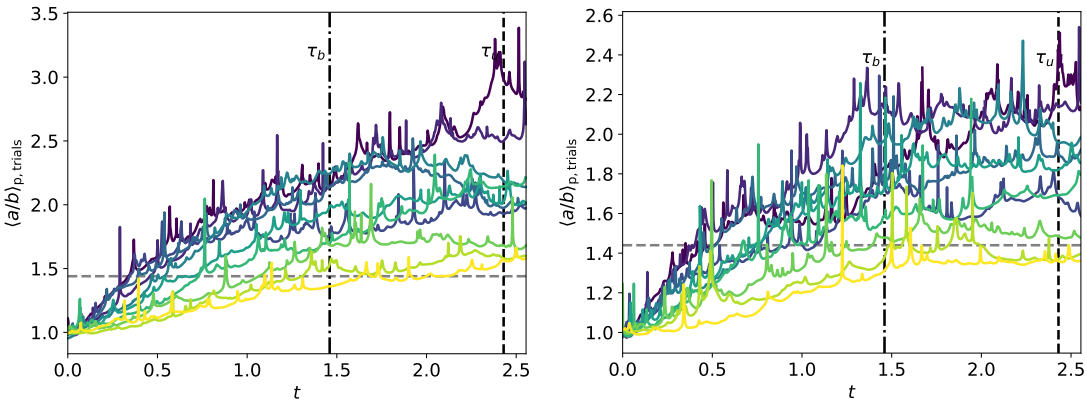
Figure 6.4: Average ratios of the pseudo-semi major axis to the pseudo-semi minor axis of a particle cloud, where the semi-major axis is with respect to the mean velocity $\mathbf{v}_0(t)$ in (a) MHD and (b) HD simulation.

dark blue lines in Figures 6.4a, 6.5a and 6.5b). After $t = \tau_u$ the ratios of all scales, except the largest, are above the threshold value for all reference directions. The differences between the reference directions are the amplitudes with decreasing scale. The smallest maximum values are reached when considering $\mathbf{v}_0(t)$ as the reference direction, where the values reach up to 2.0 (see Figure 6.4a). For the Elsässer field $\mathbf{z}_0^+(t)$ the maximum is approximately 2.2. The highest ratios, approximately 3.0, are reached when the magnetic field is the reference direction (see Figure 6.5a). The MHD flow seems to be structured with respect to these three directions in contrast to HD, where no anisotropy can be observed with respect to the mean velocity. The three (or four when including $\mathbf{z}_0^-(t)$) directions clearly seem to be relevant in the dispersion of particles in the MHD flow. When choosing a random direction this anisotropy is not observed. It remains unclear, however, how the particle dispersion is connected to the mean directions. Therefore, we want to analyze the fluctuation amplitudes and possible shear with respect to these reference directions in the following.

6.5 Large scale shear or fluctuation anisotropy?

The analysis of Lagrangian trajectories from a cloud of particles has revealed anisotropic dispersion of such particles in MHD and HD turbulence. Anisotropy of the fluctuations or shear in the velocity field, magnetic field or Elsässer variables could be the driving cause for the anisotropic dispersion.

For each cloud the velocity fluctuations with respect to a mean field $\mathbf{f}_0(t)$, $\mathbf{f} \in \{\mathbf{v}, \mathbf{b}, \mathbf{z}^\pm\}$, can be computed. The mean fields are computed as an average over all points in the cloud at each time step, analogous to equation (4.30). The velocity $\mathbf{v}_i(\mathbf{x}_i, t)$ of the i-th particle is then projected onto $\mathbf{f}_0(t)$, so that $\mathbf{v}_{i,\parallel} \cdot \mathbf{f}_0/(\|\mathbf{v}_{i,\parallel}\|\|\mathbf{f}_0\|) = \pm 1$ and $\mathbf{v}_{i,\perp} \cdot \mathbf{f}_0 = 0$.



(a) **(b)**

Figure 6.5: Average ratios of the pseudo-semi major axis to the pseudo-semi minor axis of a particle cloud in MHD, where the semi-major axis is with respect to (a) the mean magnetic field $\mathbf{B}_0(t)$ and (b) the mean positiv Elsässer variable $\mathbf{z}_0^+(t)$.

Similarly the mean velocity $\mathbf{v}_0$ is projected onto the mean field $\mathbf{f}_0$. The parallel and perpendicular velocity fluctuations are then given by

$$\delta\mathbf{v}_{i,\parallel}(t) = \mathbf{v}_{i,\parallel}(t) - \mathbf{v}_{0,\parallel}(t) \quad \text{and} \quad \delta\mathbf{v}_{i,\perp}(t) = \mathbf{v}_{i,\perp}(t) - \mathbf{v}_{0,\perp}(t), \tag{6.2}$$

where $\mathbf{v}_{0,\perp} = \mathbf{0}$, if $\mathbf{f}_0 = \mathbf{v}_0$.

To differentiate between fluctuation anisotropy and shear of the large scale flow field the rms values of the fluctuations are computed in different ways. Anisotropy of the fluctuations in the Lagrangian frame is isolated by comparing the rms of the perpendicular fluctuations to the rms of the parallel fluctuations along the Lagrangian trajectory. Thus, the temporal rms of the parallel velocity fluctuations of the i-th particle is calculated by

$$\mathrm{rms}_t(\delta\mathbf{v}_{i,\parallel}) = \sqrt{\frac{1}{T}\int_0^T \|\delta\mathbf{v}_{i,\parallel}(t)\|^2 dt}, \tag{6.3}$$

and analogously for the perpendicular fluctuations. The perpendicular rms is normalized by a factor of $\sqrt{2}$, which naturally occurs when the amplitudes of the fluctuation components $\delta v_\parallel$, $\delta v_{\perp,1}$ and $\delta v_{\perp,2}$ are subject to the same underlying distribution function $P(\mu,\sigma)$. The rms of the perpendicular fluctuation vector is then calculated by

$$\mathrm{rms}_t(\delta\mathbf{v}_{i,\perp}) = \sqrt{\frac{1}{T}\int_0^T \|\delta\mathbf{v} \perp\|^2 dt} = \sqrt{\frac{1}{T}\int_0^T \delta v_{\perp,1}^2 + \delta v_{\perp,2}^2 dt} \tag{6.4}$$

$$= \sqrt{\frac{1}{T}\int_0^T 2\left(P(\mu,\sigma)\right)^2 dt} = \sqrt{2}\mathrm{rms}_t(\delta\mathbf{v}_{i,\parallel}), \tag{6.5}$$

giving rise to a factor of $\sqrt{2}$ for isotropically distributed fluctuations. Therefore, we account for this factor in the analysis. The ensemble average is taken to be the average

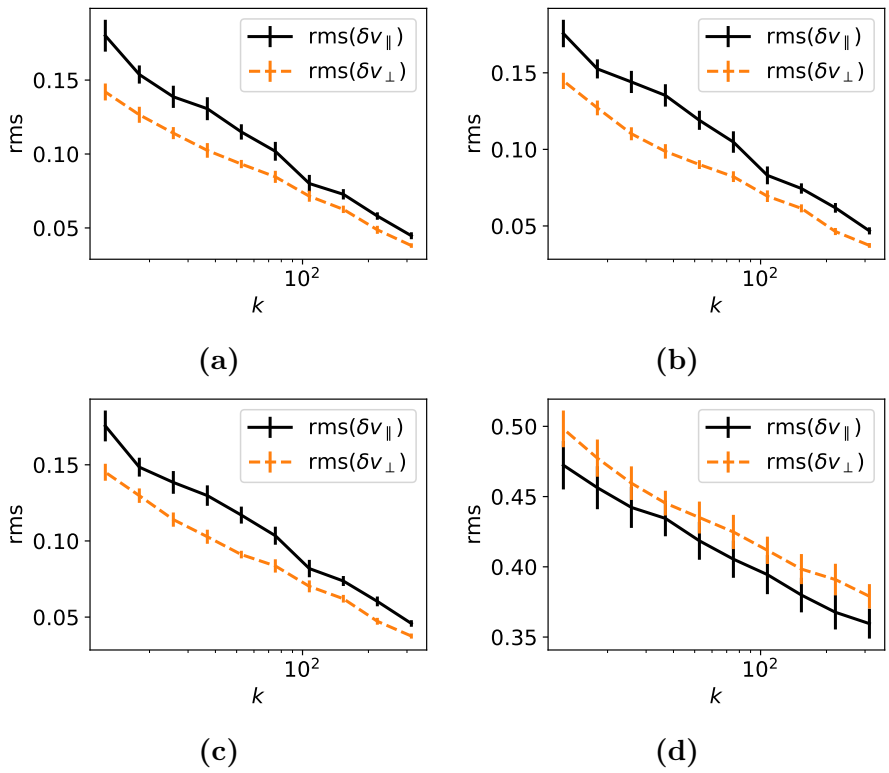


Figure 6.6: Average rms values of parallel Lagrangian fluctuations $\delta\mathbf{v}_\parallel = \mathbf{v}_\parallel - \mathbf{v}_{0,\parallel}$ and perpendicular Lagrangian fluctuations $\delta\mathbf{v}_\perp = \mathbf{v}_\perp - \mathbf{v}_{0,\perp}$ as a function of wave number k. The rms of the fluctuations is taken along each particle trajectory in time. Averages are taken over all particles in all trial clouds. The parallel and perpendicular directions are with respect to (a) the mean velocity field $\mathbf{v}_0(k,t)$, (b) the mean magnetic field $\mathbf{b}_0(k,t)$,(c) the positive Elsässer field $\mathbf{z}_0^+(k,t)$. and (d) the HD case.

over all sample particles in all clouds of a certain scale. The results are presented in Figure 6.6.

The rms of both parallel and perpendicular fluctuations decreases with increasing wave number for every mean field $\mathbf{f}_0$ in MHD and HD. The turbulent cascade where fluctuation amplitudes decrease with decreasing eddy scale could be an explanation for this. In all cases the mean rms values are linearly proportional to $-\log(k)$, which could reflect self-similarity of the turbulence. Most notably is the dominance of parallel fluctuations over perpendicular fluctuations at all scales for all mean field directions for MHD. The relative fluctuations of the particles parallel to the mean field direction is higher than perpendicular, which agrees with the anisotropic dispersion observed by CHA, PCA and pseudo-spheroid analysis. However, the difference between the rms values parallel and perpendicular is larger for smaller wave numbers corresponding to larger scales. From the anisotropy of the dispersion, which increases with wave number, the opposite is expected. The anisotropy of Lagrangian fluctuations seems to be insufficient to explain this scale-dependent anisotropic dispersion.

For HD the difference between the parallel and perpendicular rms of the fluctuations is insignificant, which is in agreement with the previous observations (compare Figures

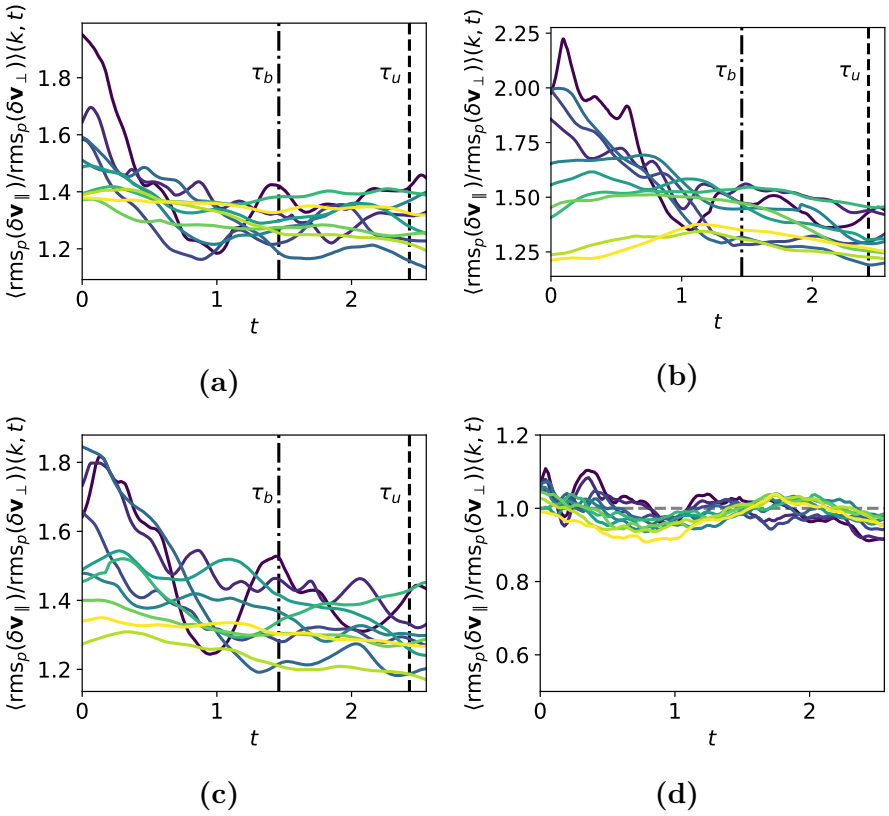


Figure 6.7: Ratios of rms values of parallel fluctuations $\delta\mathbf{v}_\parallel = \mathbf{v}_\parallel - \mathbf{v}_{0,\parallel}$ and perpendicular fluctuations $\delta\mathbf{v}_\perp = \mathbf{v}_\perp - \mathbf{v}_{0,\perp}$ as a function of time. The rms of the fluctuations is calculated between the particles in the cloud at each time step. Averages are taken over all trial clouds. The parallel and perpendicular directions are with respect to (a) the mean velocity field $\mathbf{v}_0(k,t)$, (b) the mean magnetic field $\mathbf{b}_0(k,t)$, (c) the positive Elsässer field $\mathbf{z}_0^+(k,t)$ and (d) the HD case.

6.6d and 6.4b). Here, the perpendicular fluctuations are slightly larger than the parallel fluctuations, but within the errors they are indistinguishable. Similar to the rms in MHD there is no scale-dependent deviation between the perpendicular and parallel fluctuations that could explain the scale-dependent anisotropic dispersion observed in CHA and PCA for HD (see Figures 6.2b and 6.3b).

Shear in the large scale flow field is another possible cause for anisotropic dispersion. To quantify the anisotropy of shear we compute the parallel and perpendicular velocity fluctuations as introduced in equation (6.2). Instead of calculating the rms of the Lagrangian fluctuations, the parallel rms is calculated between the particles of one cloud at each time step by

$$\mathrm{rms}_p(\delta\mathbf{v}_\parallel(t)) = \sqrt{\frac{1}{N}\sum_{i=1}^{N} \|\delta\mathbf{v}_{i,\parallel}(t)\|^2}. \tag{6.6}$$

Further, we calculate the ratio of $\mathrm{rms}_p(\delta\mathbf{v}_\parallel(t))$ to $\mathrm{rms}_p(\delta\mathbf{v}_\perp(t))$ to quantify which direction experiences more shear. Again, we account for the factor of $\sqrt{2}$ of the perpendicular rms. The ensemble average can be calculated over all clouds of a specific scale. The ratio is thus a function of scale and time. We calculate the ratios of the parallel to the perpendicular rms for the mean directions $\mathbf{v}_0(t)$, $\mathbf{b}_0(t)$ and $\mathbf{z}_0^+(t)$ in MHD and for $\mathbf{v}_0(t)$ for HD. The

results are presented in Figure 6.7. The scales are again indicated by colors corresponding to the wave numbers depicted in Figure 6.2b. For MHD the ratio is scale-dependent at the initial condition at $t = 0$, where small scales show a higher ratio than large scales for all mean directions. The small scales show a ratio between 1.5 and 2.2, whereas the large scales show a ratio between 1.2 and 1.4, i.e. the rms of the parallel fluctuations is larger than the rms of the perpendicular fluctuations for all scales and is increasingly larger towards small scales. This is consistent with the observed anisotropic dispersion, where small scales disperse more anisotropically than large scales. For fluctuations projected onto the mean velocity the scale-dependence is lost after a time of $t \approx 0.5$ as the ratios of the small scales decrease to below 1.5 (see Figure 6.7a). The ratios of the large scales are approximately constant around 1.3-1.4 for all times. At times larger than 0.5 there is no scale-dependence, but the ratios of all scales are between 1.2 and 1.45, so that the rms of $\delta\mathbf{v}_{\parallel}$ still dominates the rms of $\delta\mathbf{v}_{\perp}$. As the rms is calculated over the particles instead of time the ratios represent anisotropy of shear in the mean velocity field. At all times the parallel shear is larger than the perpendicular shear. At early times the ratio of the shear parallel and perpendicular to the mean flow velocity is scale-dependent, where the smallest scales show the largest ratios. For $\mathbf{b}_0$ and $\mathbf{z}_0^+$ we also observe scale-dependence at $t = 0$ with higher ratios for smaller scales. Also the ratios decay in a similar manner as for the mean velocity. However, the scale-dependence vanishes at later times, namely around $t \approx 1$ for the mean magnetic field (see Figure 6.7b) and around $t \approx 0.8$ for the mean Elsässer variable (see Figure 6.7c). The ratio is highest for the magnetic field with values up to 2.2 for the smallest scales, consistent with the anisotropy of the dispersion, where the highest anisotropy is found with the mean magnetic field as the reference direction (see Figure 6.5a). For later times they show no scale-dependence and vary around 1.2 and 1.5 for both $\mathbf{b}_0$ and $\mathbf{z}_0^+$. The transition from scale-dependent anisotropic shear to scale-independent anisotropic shear is observed for all mean directions and seems to be a property of the MHD flow. The transition approximately agrees with the saturation of anisotropy measures introduced in the previous sections. The ratio of the convex hull area to its volume saturates around $t \approx 0.7$, which is comparable to the loss of scale-dependence in shear with respect to the mean Elsässer variable (compare Figures 6.2a and 6.7c). The PCA and pseudo-spheroid analysis do not necessarily saturate, however, the decreasing increase of the anisotropy measures is in agreement with the loss of scale-dependence in shear anisotropy over time. These changes in the scale-dependence of the shear anisotropy possibly correspond to changes between an advective transport to a transport similar to an isotropic diffusion process at later times.

A very different observation is made for HD, shown in Figure 6.7d. The rms of the parallel fluctuations is approximately equal to the rms of the perpendicular fluctuations at all times and scales. The mean flow velocity does not seem to be a direction of reference for anisotropic dispersion in HD, which agrees with the observations of the pseudo-spheroid analysis, where we could not detect anisotropic dispersion along the mean flow velocity

(see Figure 6.4b). We are unable to isolate a reference direction for the anisotropic dispersion observed by CHA and PCA for HD.

6.6 Discussion

The dispersion of particles was investigated by previously used as well as newly introduced methods. The classical displacement of particle pairs was calculated for clouds of particles of different initial scale. The mean squared relative displacement scales with t^{-2}, which suggests that the particles are advected rather than diffusing isotropically (Batchelor, 1950). Previous findings of two-particle dispersions in MHD are consistent with our findings (Busse et al., 2007). No change in the slope of the mean squared relative displacement is observed, in contrast to Busse et al. (2007), which could be accounted for by the short simulation time which only reaches up to one large eddy turnover time. The particle displacement does not give any insight into anisotropy or directional preference of the dispersion.

The analysis of the convex hull of the particle cloud has been introduced by Pratt et al. (2017) and was applied to the particle clouds here. When taking the ratio of the area to volume$^{2/3}$ of the convex hull we observe a significant anisotropy, meaning deviation from the isotropic dispersion, in agreement with Pratt et al. (2017). The ratio is scale-dependently increasing and saturates for times later than the magnetic large eddy turnover time. The smallest scales reach up to higher ratios, indicating stronger anisotropy than the larger scales. In HD also a scale-dependent increase is observed which peaks at around $\tau_u/6$ and after that continuously decays. At τ_u no scale dependence is observed and the ratios keep decaying towards the isotropic ratio. Large scale anisotropic shear at the initial clouds transitioning to isotropic dispersion/diffusion could explain this behaviour. A significantly different behaviour is found in MHD where a decay of the ratios is absent and at the kinetic large eddy turnover time the scale-dependent anisotropy remains. This longer anisotropy in MHD is consistent with Busse et al. (2007), who explain the slowed down relative dispersion by a constricting effect of the local mean magnetic field.

In addition to the CHA the newly introduced PCA was applied and showed similar results to the CHA. In addition to the CHA the PCA could give an insight into the shape of the particle clouds. Whether the clouds take a discus shape or a cigar shape can be tested by the PCA when taking the ratio of all three principal components of the covariances of the particle distances into account. The analysis did not reveal any preferred shape, not confirming a possible cigar shape. A cigar shape would be expected from the wave vector anisotropy found in, for example, the critical balance theory (Goldreich and Sridhar, 1995). The distribution of energy in **k**-space does not directly infer the shapes of dispersing clouds in the physical space. There was no correlation found between the direction of highest variance and the mean flow or mean magnetic field direction, as well as no dependence of anisotropy on the initial cross-helicity.

Both CHA and PCA revealed anisotropy in the particle dispersion with significant differences between MHD and HD. The pseudo-spheroid analysis was applied to the particle clouds to test any anisotropy with respect to the characteristic mean fields found in the system. For MHD, anisotropy is found for all mean field directions, namely the mean velocity, the Elsässer variables and the mean magnetic field, the latter showing the highest anisotropy. The pseudo-spheroid analysis is in agreement with the CHA and the PCA in MHD, showing scale-dependent increase of anisotropy. The mean velocity and the mean magnetic field seem to be relevant reference directions in the anisotropic dispersion of particles. The pseudo-spheroid analysis does not show any anisotropy for HD with respect to the mean flow velocity. Deardorff and Peskin (1970) and Shen and Yeung (1997) find faster dispersion in HD shear-flow if the initial displacement is in the mean flow direction. The mean flow direction does not seem to be relevant in isotropic turbulence but when a strong large scale shear is induced.

An explanation for the anisotropic dispersion was given by the analysis of Lagrangian fluctuations and the mean field shear of the velocity. The Lagrangian fluctuations were analyzed with respect to the characteristic mean fields. For MHD we find that parallel Lagrangian velocity fluctuations are slightly larger than the perpendicular fluctuations at all scales and all mean fields. In contrast, fluctuations parallel and perpendicular to the mean flow velocity are approximately equal in HD. This is consistent with the isotropic dispersion (with respect to the mean flow velocity) observed by the pseudo-spheroid analysis. The ratio of the Lagrangian fluctuations parallel and perpendicular is only slightly scale-dependent. Large scales show a higher discrepancy between perpendicular and parallel fluctuations than small scales. The scale-dependent anisotropic dispersion, where small scales show higher anisotropy, appears to have another origin. Therefore, the shear of the mean field was analyzed for the particle clouds. For MHD the ratios of the initial shear parallel to the shear perpendicular is scale-dependent with larger ratios for small scales than large scales. All scales show a larger shear parallel than perpendicular to the mean fields, in agreement with the anisotropic dispersion observed in CHA, PCA and pseudo-spheroid analysis. The decay of scale-dependence in the shear approximately coincides with the loss of increase of the anisotropy in all the methods. The influence of larger scale shear on smaller scale clouds seems to vanish. The shear of the velocity fluctuations appears to be the main driver of anisotropy in the MHD particle dispersion and seems to be governed by the local mean magnetic field and the local mean flow velocity or the Elsässer variable as the combination of both. Busse et al. (2007) consistently find anisotropy with respect to the local mean magnetic field in two-particle dispersion. In HD the shear with respect to the mean flow velocity seems to be isotropic, similar to the Lagrangian fluctuations. Thus we are unable to explain the anisotropy in the dispersion observed by CHA and PCA in HD. The direction of reference does not seem to be the mean flow velocity.

7 Summary & outlook

In the present thesis we have investigated several properties of MHD flow fields. The large scale influence of forcing on the velocity and the magnetic field in an incompressible MHD simulation was investigated. The dominance of this forcing in the correlations of the fields and the crosscorrelation between those fields was shown. The forcing only acts on the velocity, which is decoupled from the magnetic field, indicating the frozen-in assumption is invalid here. To analyze the processes of the turbulent flow decoupled from the forcing several methods have been introduced. The scale-dependent Eulerian correlation function allows the study of temporal dynamics at intermediate scales without the influence of the constant forcing at the largest scales. The decorrelation times as functions of scale are not in agreement with one characteristic time of the system, but could be explained by a superposition of them. Lagrangian trajectories of particles were extracted and besides the correlations of the fluctuations, the anisotropic properties of the dispersion have been analyzed. Techniques of classical two particles displacements were applied to the clouds and were shown to be in agreement with previous findings, where a scaling of t^2 is observed (Busse et al., 2007). This is an indication for an advective/dispersive process rather than a diffusive process (Batchelor, 1950; Busse et al., 2007). Several new methods were applied to particle clouds to further investigate the anisotropy of the dispersion. The convex hull analysis (CHA) and the newly introduced principal component analysis (PCA) show similar anisotropy in the dispersion, which differs significantly between MHD and HD turbulence. Using the a-priori knowledge of characteristic mean field directions in the flow the novel pseudo-spheroid analysis underlined the anisotropy found for MHD in CHA and PCA, but failed to do so for HD. The HD dispersion shows anisotropy, but the direction along which this anisotropy occurs does not seem to be the mean flow direction. An explanation for the anisotropic dispersion in MHD was found in the anisotropy of fluctuation amplitudes and, more importantly, the scale-dependent shear in the mean fields. Large scale shear and fluctuations being the main driver in the anisotropy of particle dispersion agrees with large scale random sweeping or advection of Alfvén waves being responsible for Eulerian decorrelations.

For future research a longer simulation time over multiple large eddy turnover times is highly desirable. Possible boundary errors or lack of statistics in the temporal domain could be excluded confidently and the determination of the frequency spectra as well as the correlation functions gets more robust. The PCA can be applied to the Lagrangian

fluctuations or the shears to determine the direction of highest shear. A comparison with the characteristic mean fields of the flow is then possible and could lead to a better understanding of the anisotropic dispersion. A thorough analysis of the averaging scales of the local mean fields with respect to the fluctuation scales, or the scales of the displacement between the dispersing particles, is also necessary. The estimation of the local mean magnetic field is found to have a large influence on anisotropy analysis in MHD turbulence (Gerick et al., 2017). The analysis of the actual flow fields is a very young field of research with many open questions. The connection between particle dispersion on different scales and the energy associated with eddies/fluctuations on this scale remains poorly understood. Linking the distribution functions of the fluctuations in both frames is highly desirable and very recent approaches in HD are advancing in that field (Kamps et al., 2009; Lalescu and Wilczek, 2017). Also, better understanding of the connection between fluctuation scale and spatial scale in the dispersion is needed. The novel tools introduced in this thesis show promising results for the investigation of anisotropy of multiple particle dispersion and find good application in this uprising field of research.

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A Appendices

A.1 Analytical derivation of Taylor-Green forcing correlation

The autocorrelation function of the forcing $\mathbf{F^u}$ in xy-planes as a function of z is given by

$$R^E_{FF}(\mathbf{r}, z, \tau)\rangle_{xy} = R^E_{FF}(\mathbf{r}, z) = \frac{\langle \mathbf{F^u}(\mathbf{x}) \cdot \mathbf{F^u}(\mathbf{x}+\mathbf{r})\rangle_{xy}}{\langle (\mathbf{F^u}(\mathbf{x}))^2\rangle_{xy}}, \tag{A.1}$$

with spatial lags only in the x and y direction, so that $\mathbf{r} = (r_x, r_y, 0)$. The numerator is calculated to be

$$\langle \mathbf{F^u}(\mathbf{x}) \cdot \mathbf{F^u}(\mathbf{x}+\mathbf{r})\rangle_{xy} = \int_0^{2\pi}\int_0^{2\pi} F^u_x(\mathbf{x})F^u_x(\mathbf{x}+\mathbf{r}) + Fyx^u(\mathbf{x})F^u_y(\mathbf{x}+\mathbf{r})dxdy \tag{A.2}$$

$$= f_0^2\cos(k_f z)^2 \int_0^{2\pi}\int_0^{2\pi} (\sin(k_f x)\cos(k_f y)\sin(k_f(x+r_x))\cos(k_f(y+r_y))$$

$$+ \cos(k_f x)\sin(k_f y)\cos(k_f(x+r_x))\sin(k_f(y+r_y)))dxdy \tag{A.3}$$

$$= 2f_0^2\pi^2\cos(k_f z)^2\cos(k_f r_x)\cos(k_f r_y), \tag{A.4}$$

and the denominator is given by

$$\langle (\mathbf{F^u}(\mathbf{x}))^2\rangle_{xy} = f_0^2\cos(k_f z)^2 \int_0^{2\pi}\int_0^{2\pi} \left(\sin(k_f x)^2\cos(k_f y)^2 + \cos(k_f x)^2\sin(k_f y)^2\right) dxdy \tag{A.5}$$

$$= 2f_0^2\pi^2\cos(k_f z)^2, \tag{A.6}$$

so that

$$R^E_{FF}(\mathbf{r}, z) = \cos(k_f r_x)\cos(k_f r_y). \tag{A.7}$$

Since the numerator and denominator are 2π-periodic functions, this is also true for an integration over $\mathbb{R}^2$ instead of the $[0, 2\pi]^2$ domain.

We can further compute the rms of this correlation. The rms of a continuous functions $f(t)$ is defined as

$$\mathrm{rms}(f) = \sqrt{\frac{1}{T}\int_0^T f(t)^2 dt}, \tag{A.8}$$

so that

$$\mathrm{rms}\left(R^E_{FF}(\mathbf{r}, z)\right) = \left(\frac{1}{(2\pi)^2}\int_{-\pi}^{\pi}\int_{-\pi}^{\pi}\cos^2(k_f r_x)\cos^2(k_f r_y) dr_x dr_y\right)^{1/2} \tag{A.9}$$

$$\overset{k_f=2}{=} \frac{1}{2}. \tag{A.10}$$

A.2 Threshold values for anisotropic particle dispersion markers

We estimate threshold values for the analysis tools introduced, namely the ratio of the area to the volume of a convex hull of particle clouds, the ratio of the singular values from principal component analysis (PCA) and the so-called spheroid analysis. To do so, we consider the initial condition of n particles equally distributed on a sphere with radius 1 and propagate each particle in time following a Brownian motion, so that

$$\mathbf{x}_i(t_{k+1}) = \mathbf{x}_i(t_k) + \mathbf{u}_p(t_k), \tag{A.11}$$

where $\mathbf{u}_p$ is a random normally distributed velocity. The particle cloud is thus dispersing isotropically in space. At each time step t_k the convex hull and the PCA are computed and the respective ratios are calculated. The average is taken from 10000 trials. The results are shown in Figure A.1 for different numbers $n \in [10, 20, 50, 100, 200, 500]$ of points on the initial sphere. On average the ratios are constant in time, after a jump from the initial sphere. Also observed is a decrease of these constant ratios when increasing the number of points. This is expected from the increasing deviation of the convex hull polygon to a sphere with decreasing points. Similarly, for the PCA a non-uniform distribution in space is expected for lower number of points. The difference between the threshold values decreases for high n and to save computational costs a number of 50 points on the sphere is taken to be sufficiently high. Thus, a threshold value of $\langle S/V^{2/3}(36\pi)^{-1/3}\rangle \approx 1.14$ is taken for the convex hull analysis and a threshold value of $\langle \sigma_1/\sigma_2\rangle \approx 1.35$ for PCA.

In case of the pseudo spheroid analysis we compute the standard deviations of the semi axes ratios. The mean ratio over all trials approaches 1 for all times in the Brownian motion trajectories, regardless of the number of points on the initial sphere. The standard deviation can be added to the mean value of 1 and used as a significance threshold. The values are shown in Figure A.1c, for 50 points the mean value plus standard deviation is

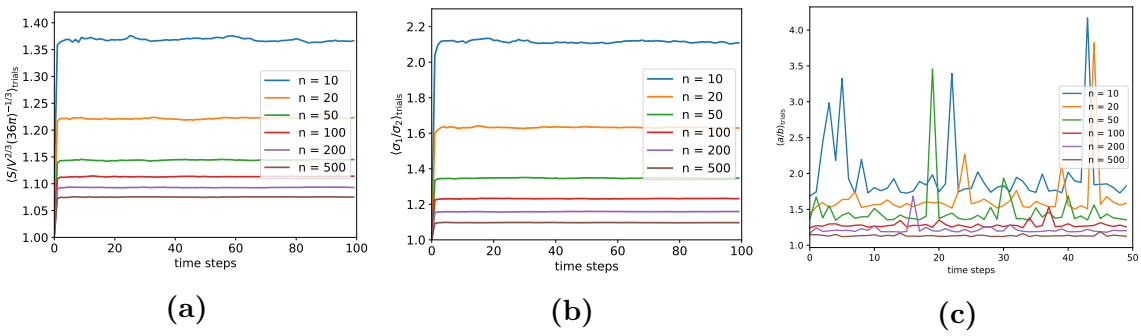

Figure A.1: (a) Normalised ratios of the area to the volume$^{2/3}$ of the convex hull and (b) ratios of the largest to the second largest singular value, σ_1/σ_2, for different numbers n of points on the initial sphere and under Brownian motion over time.

$\mu + \sigma \approx 1.44$, which is chosen as the threshold value. Although 2×10^5 trials for each n were calculated the standard deviations are not constant in time, which shows the large amount of trials needed to get sufficient statistics for this ratio.

Eidesstattliche Erklärung

Hiermit versichere ich, Felix Gerick, dass die vorliegende Arbeit von mir selbständig ohne fremde Hilfe verfasst wurde, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt und Zitate kenntlich gemacht wurden. Ich versichere auch, dass die von mir eingereichte schriftliche Version mit der digitalen Version übereinstimmt. Weiterhin erkläre ich, dass die Arbeit in gleicher oder ähnlicher Form noch nicht Bestandteil einer Studien- oder Prüfungsleistung war.

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